Building AI Agents with Multimodal Models

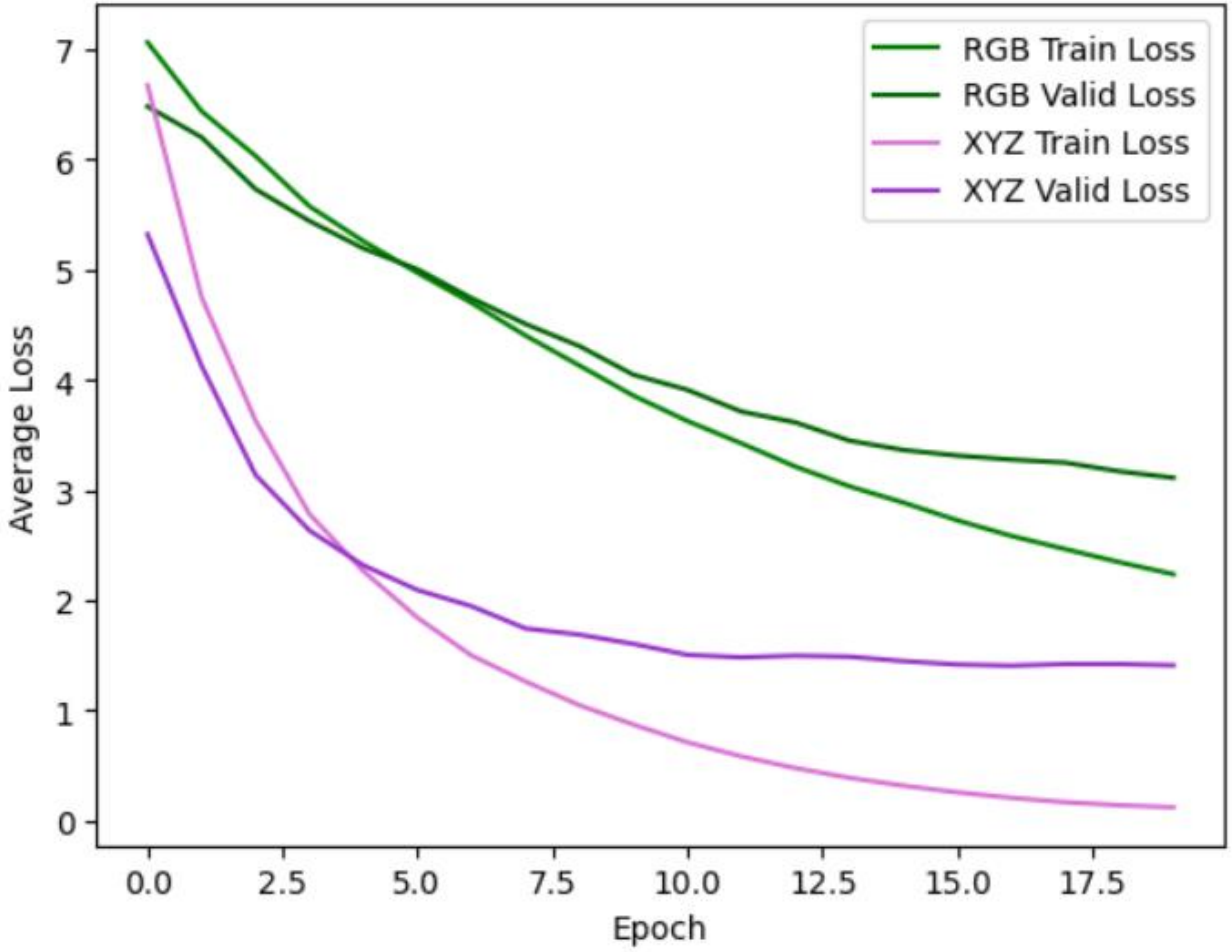
Part 2: Intermediate Fusion and Contrastive Pretraining

Agenda

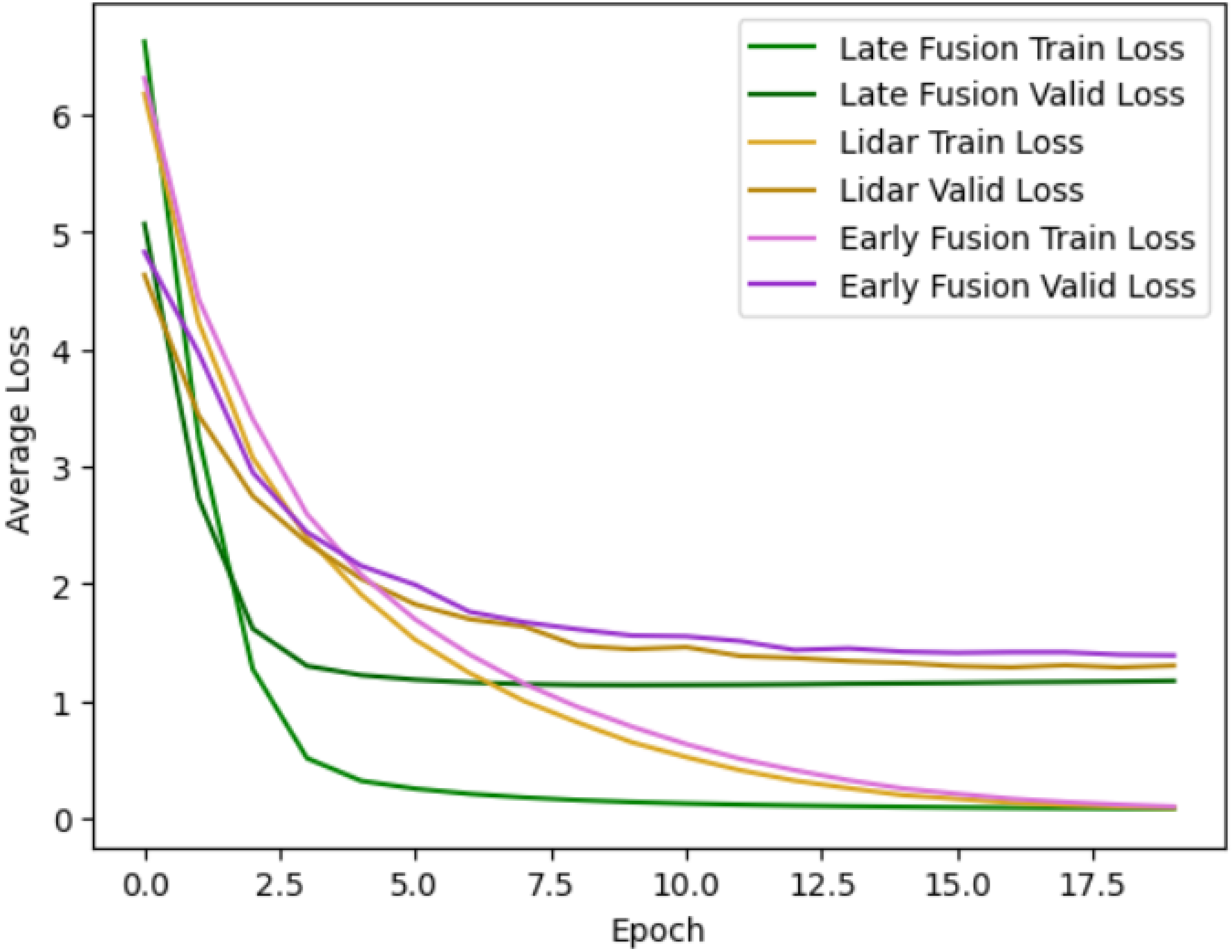
- Part 1: Multimodal Data
- Part 2: Intermediate Fusion
- Part 3: Cross-modal Projection and OCR Pipelines
- Part 4: VSS and GraphRAG
- Part 5: Wrap-up & Assessment

Experiment Results

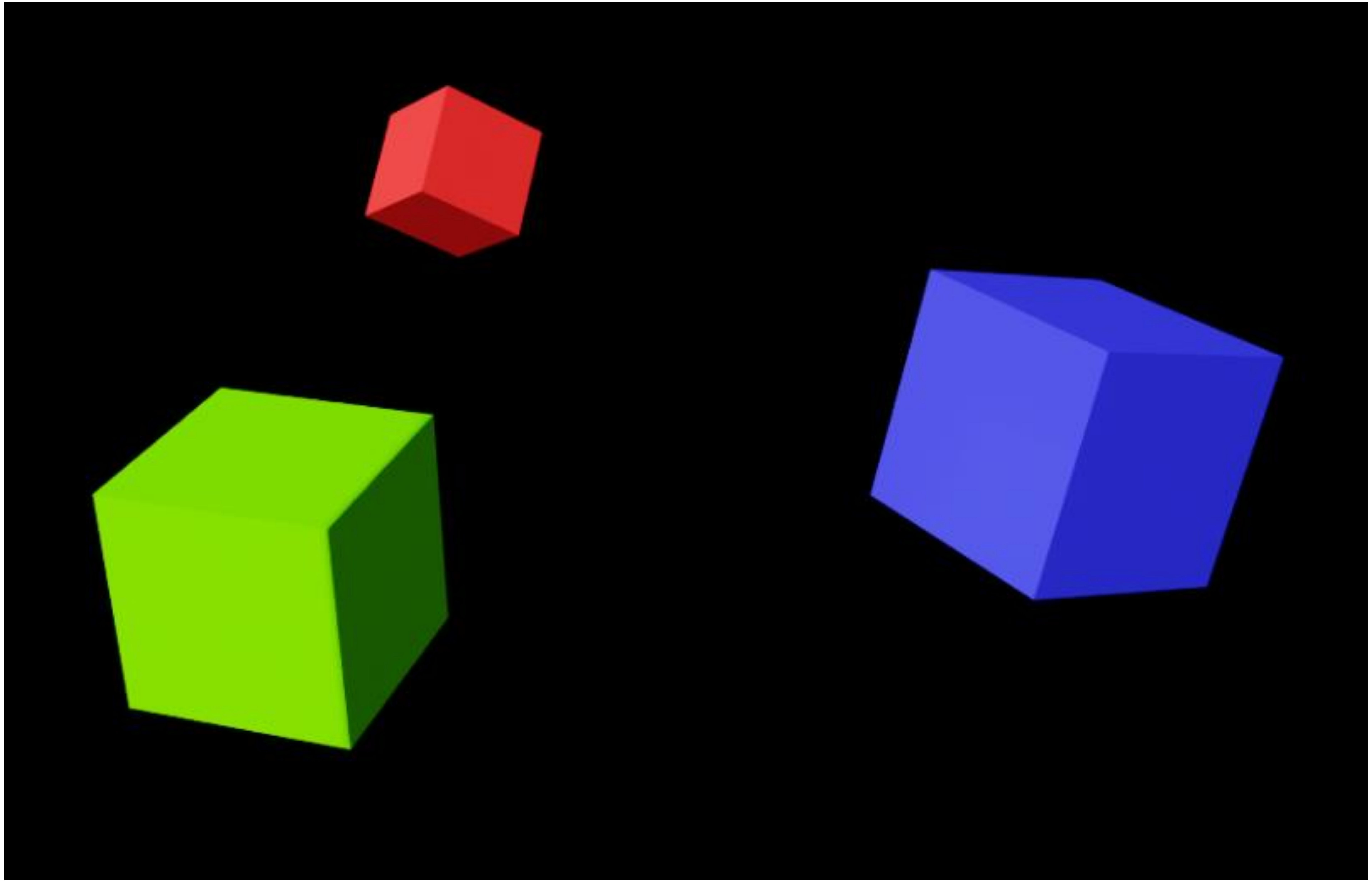
Single Modal Data - Results



Multimodal Data - Results



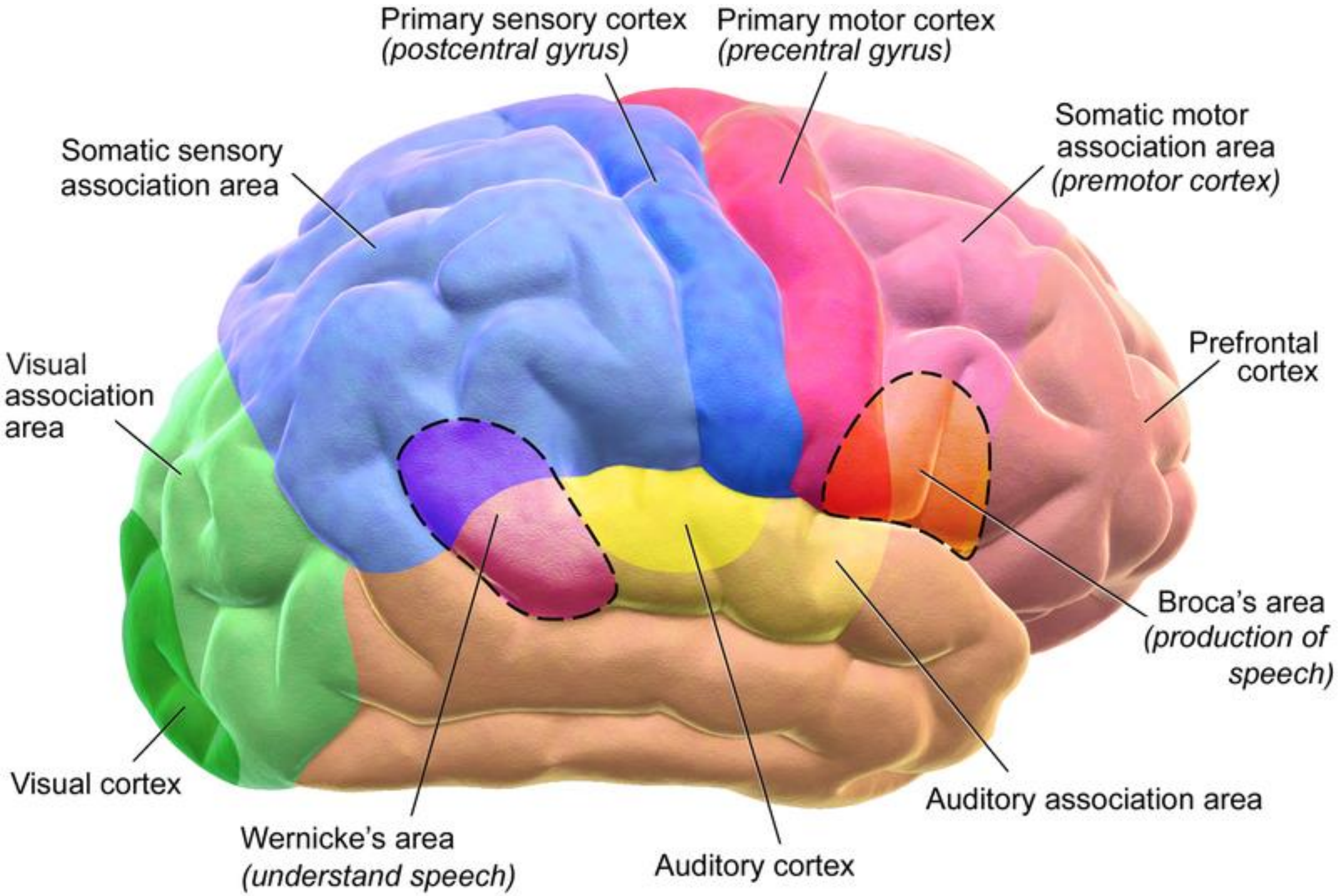
Next Experiment



Lab 2a - Intermediate Fusion

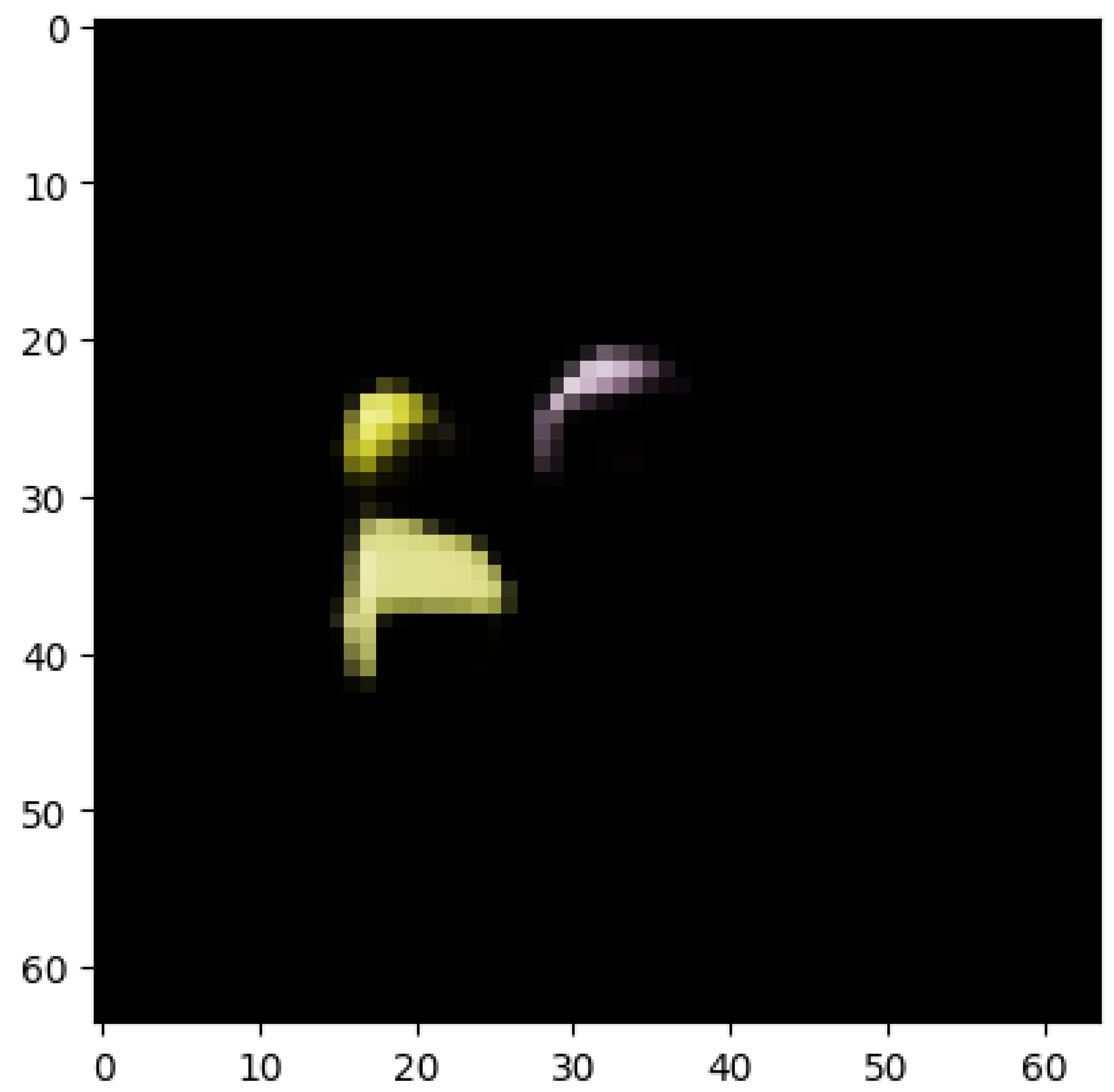
Human Inputs and Outputs

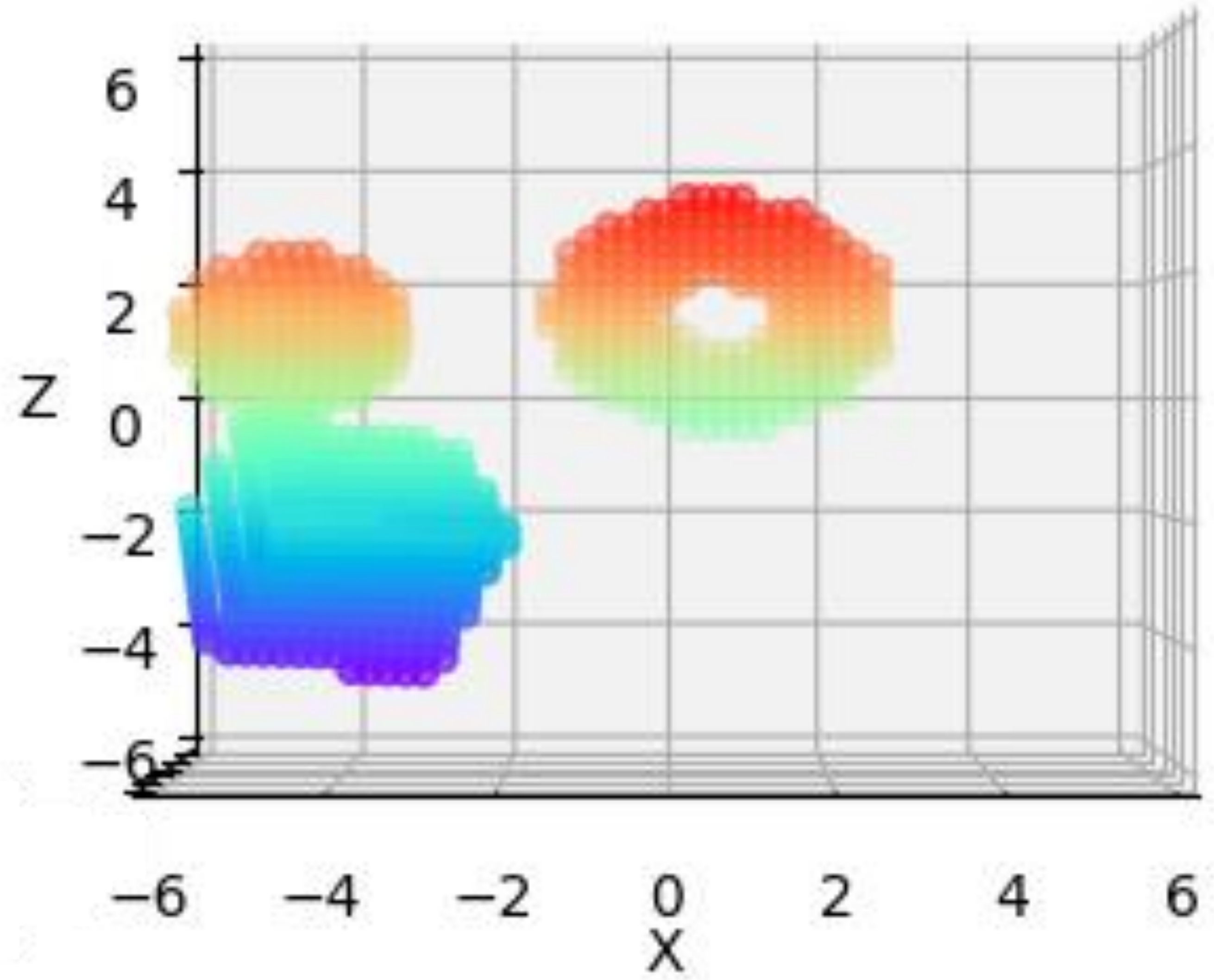
The Human Brain as a Multimodal Model



Blausen.com staff (2014). "Medical gallery of Blausen Medical 2014". *WikiJournal of Medicine* **1** (2). DOI:10.15347/wjm/2014.010. ISSN 2002-4436

Intermediate Fusion





Convolution

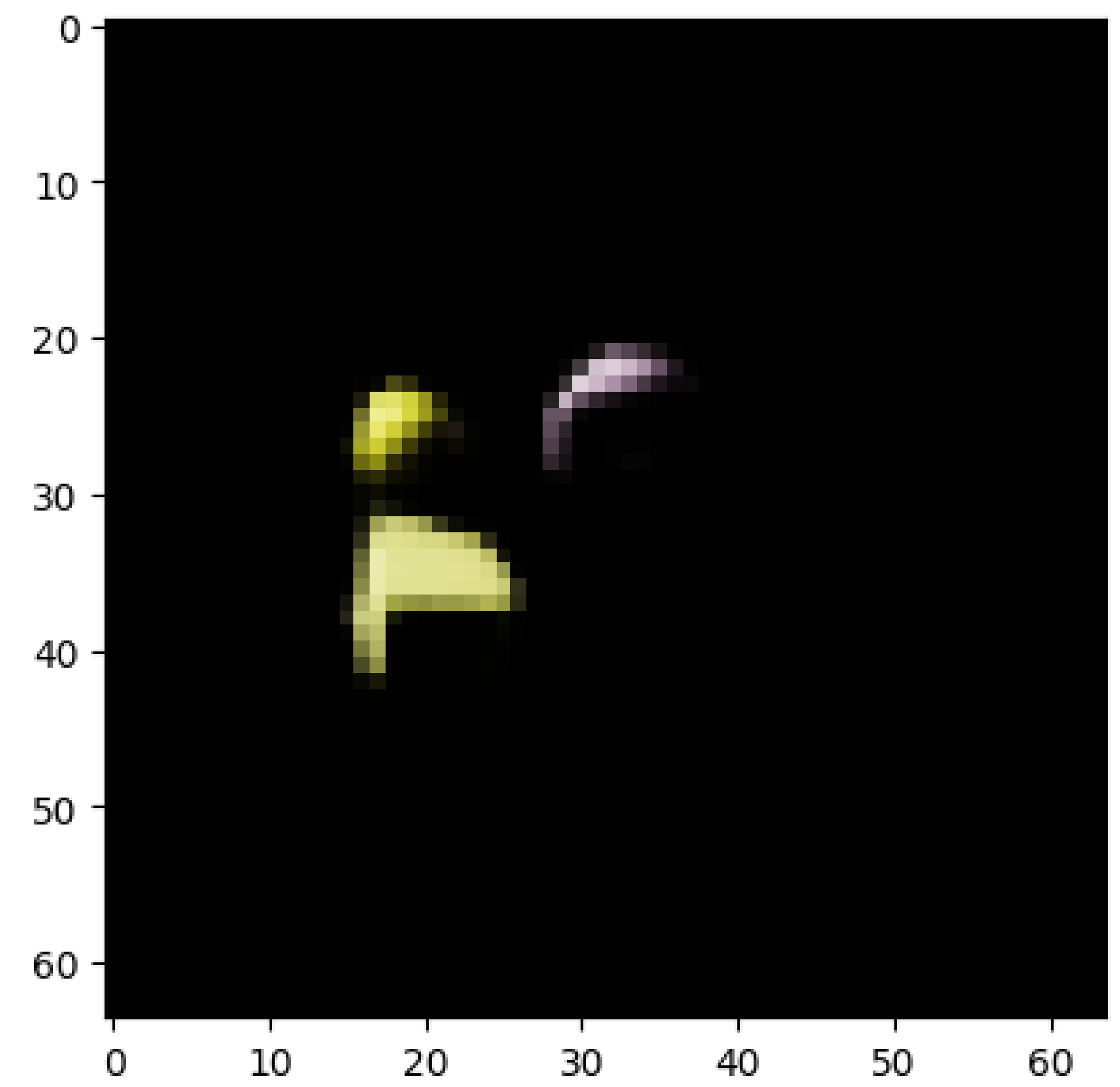
Flatten

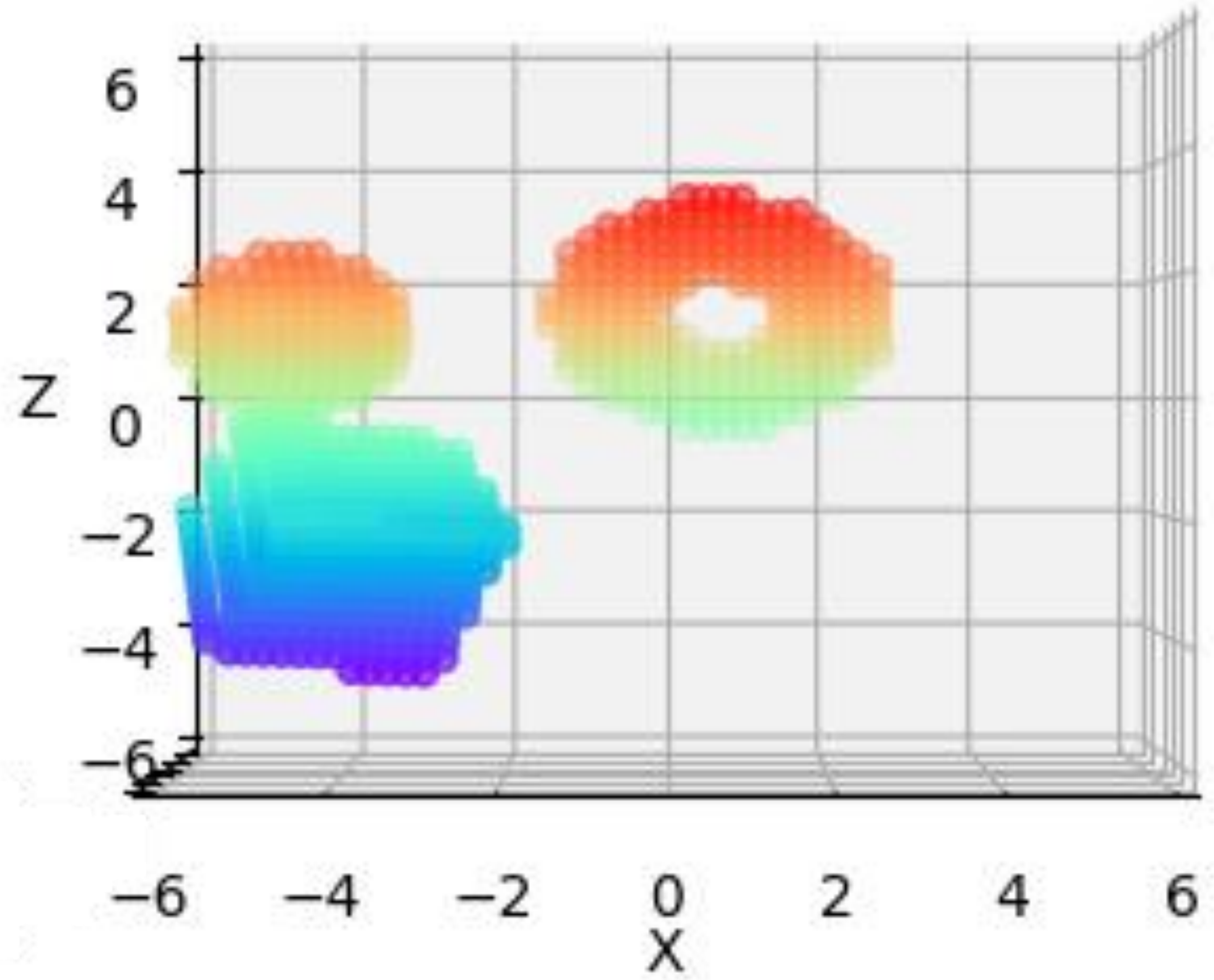
Convolution

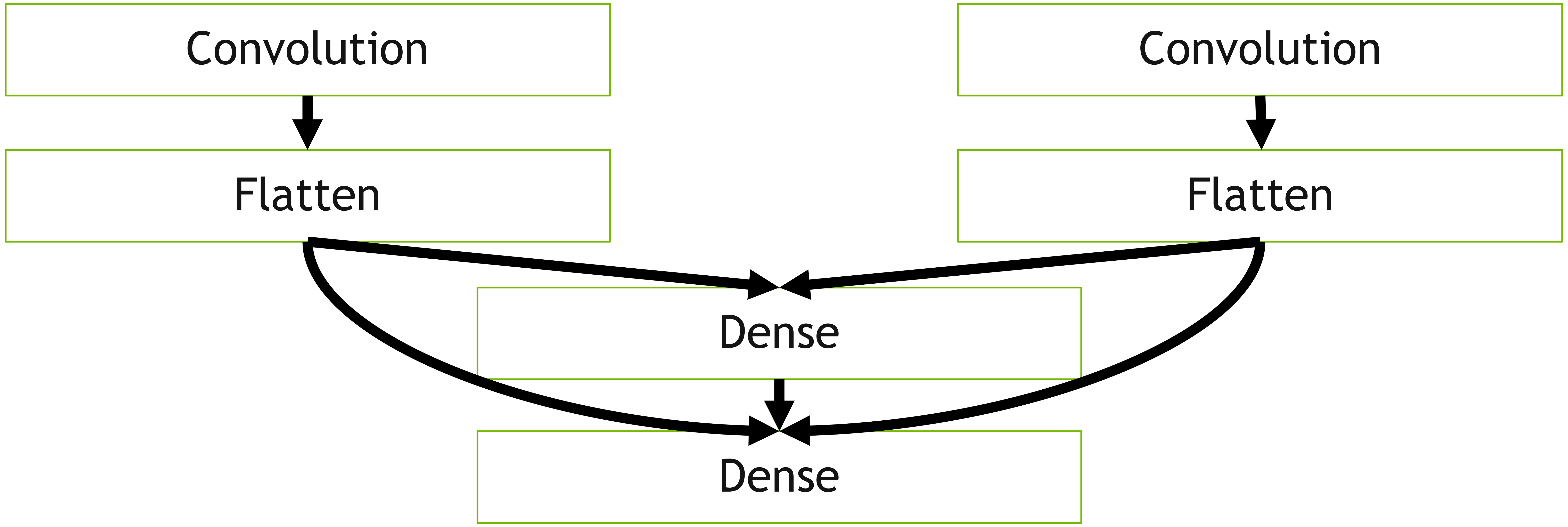
Flatten

Dense

Intermediate Fusion

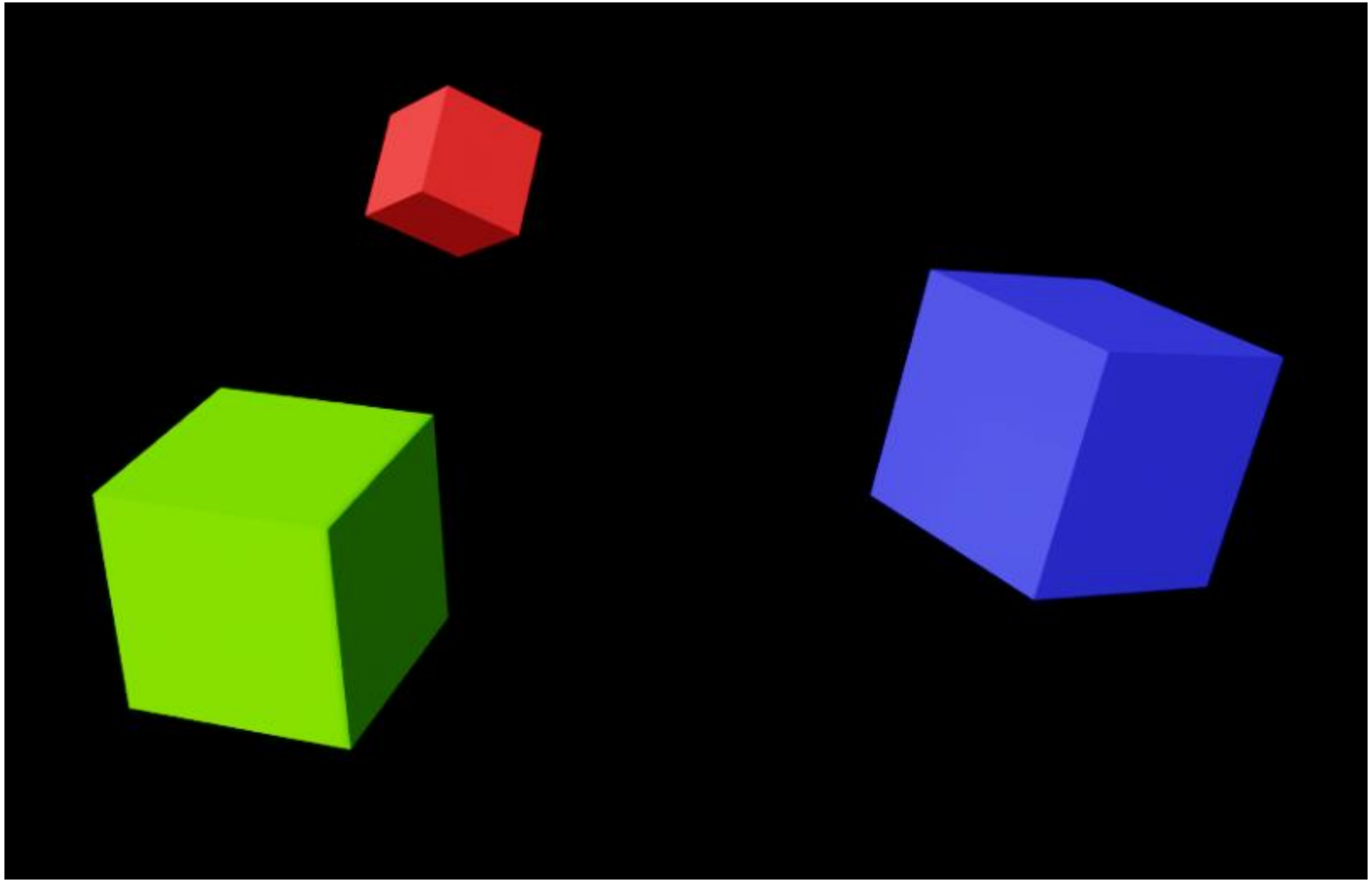






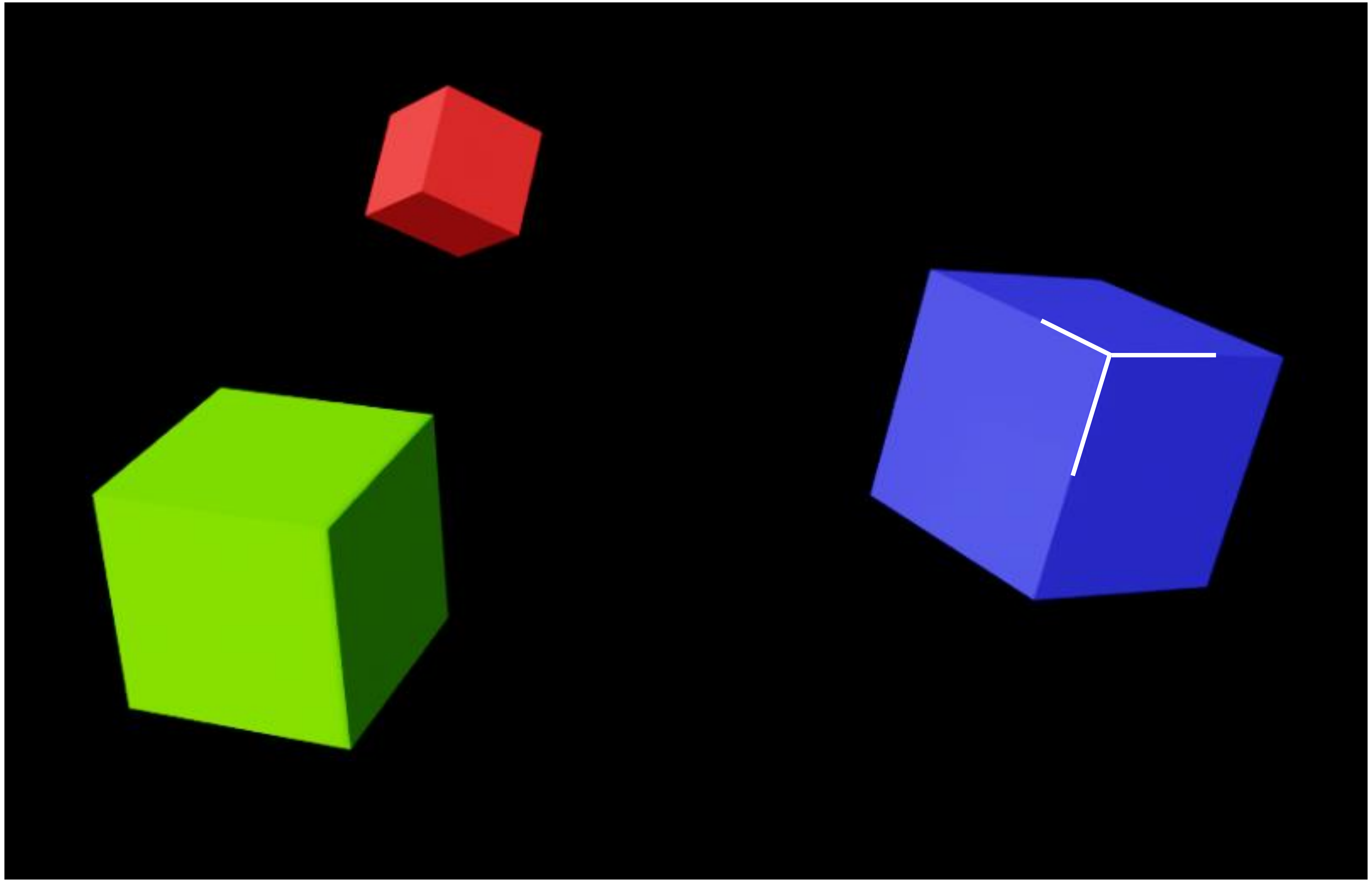
Intermediate Fusion

Finding the Center



Intermediate Fusion

Finding the Center



Intermediate Fusion

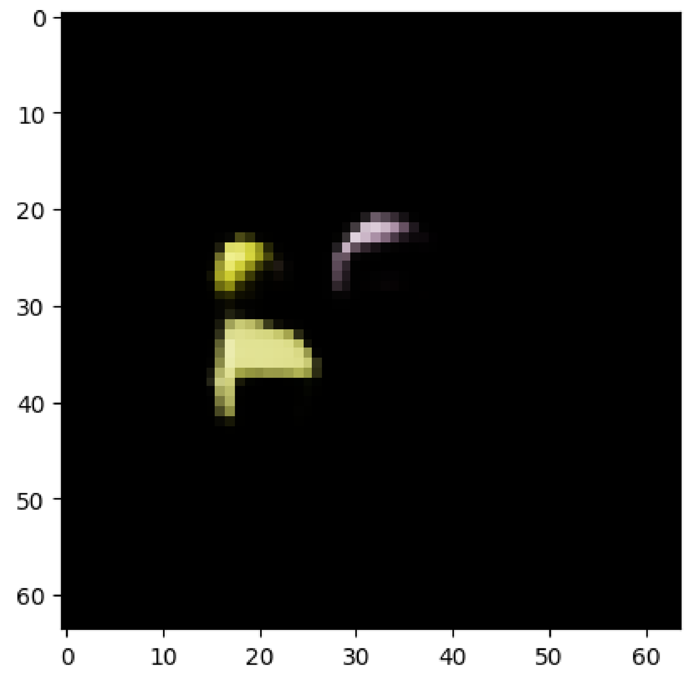
Finding the Center

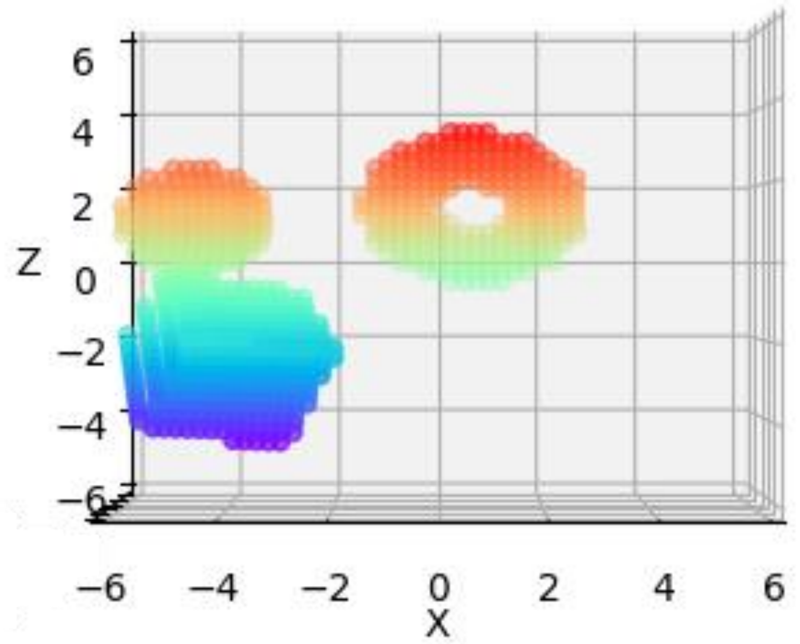
Static after training

$$y = mx + b$$

Variable after training is complete

Intermediate Fusion





Convolution → Convolution → Flatten

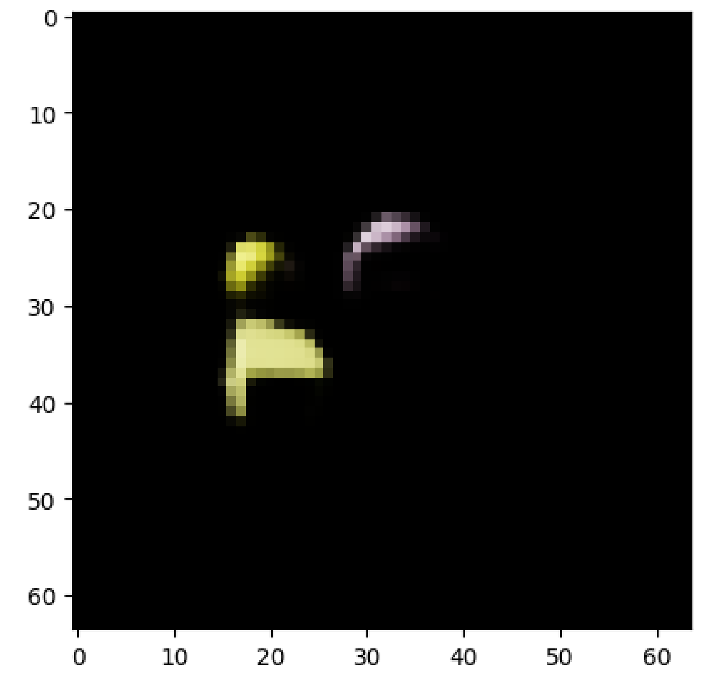
Convolution → Convolution → Flatten

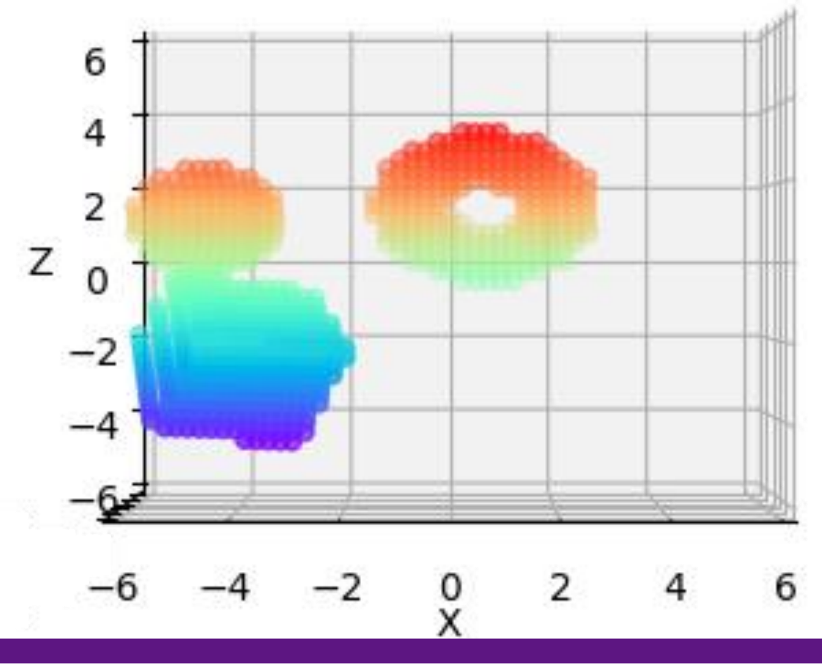
Concatenate

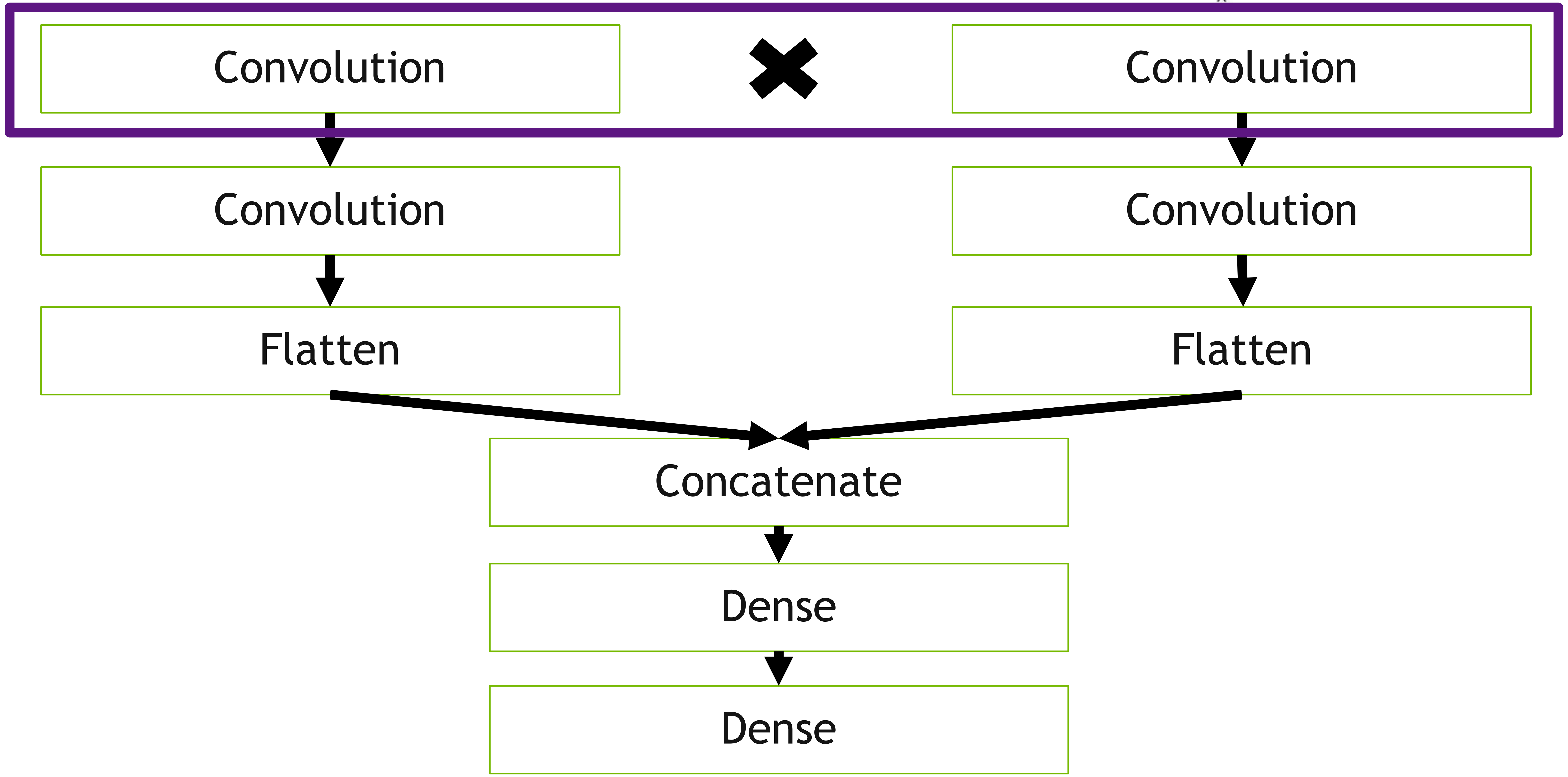
Dense

Dense

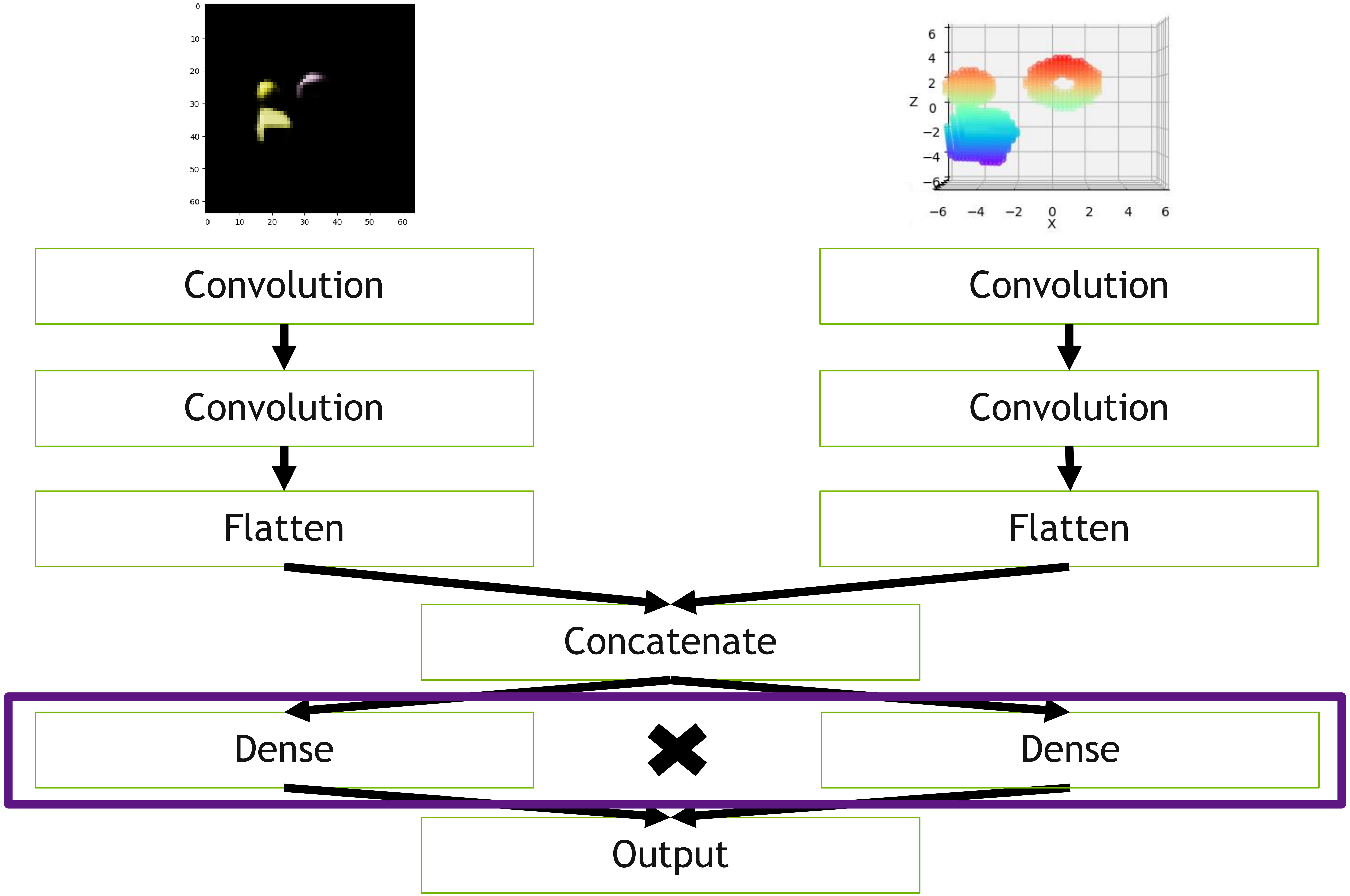
Intermediate Fusion



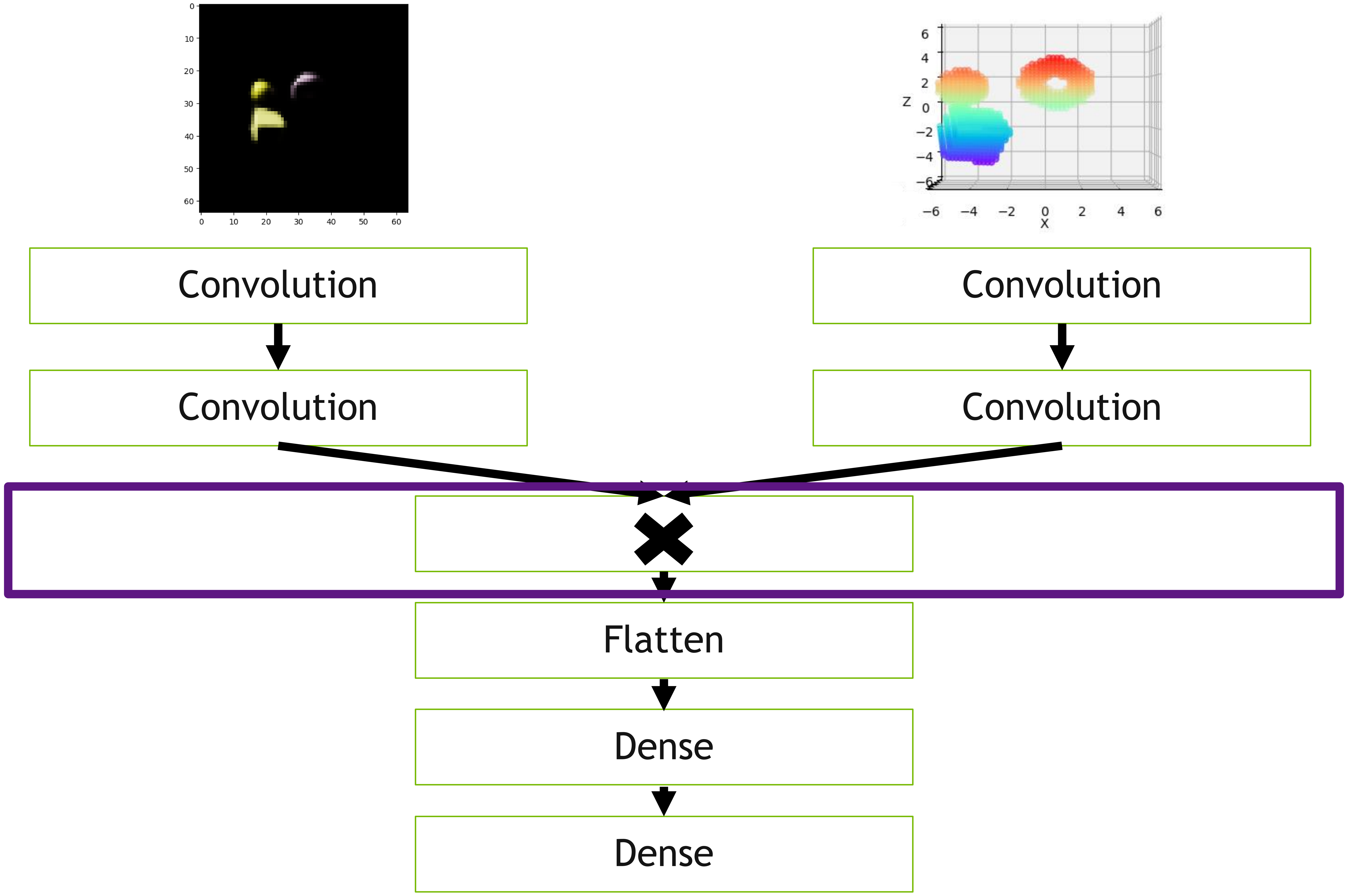




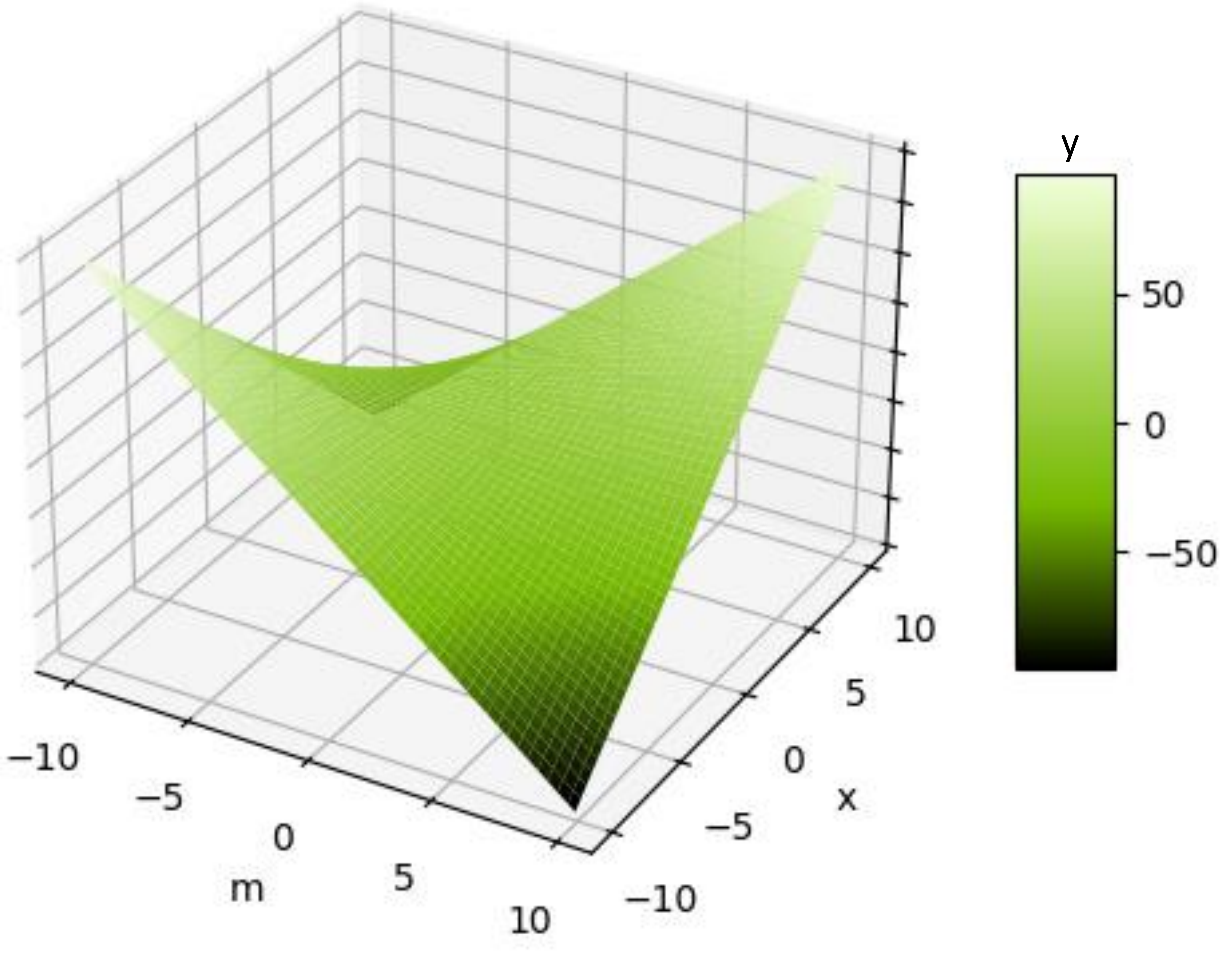
Intermediate Fusion



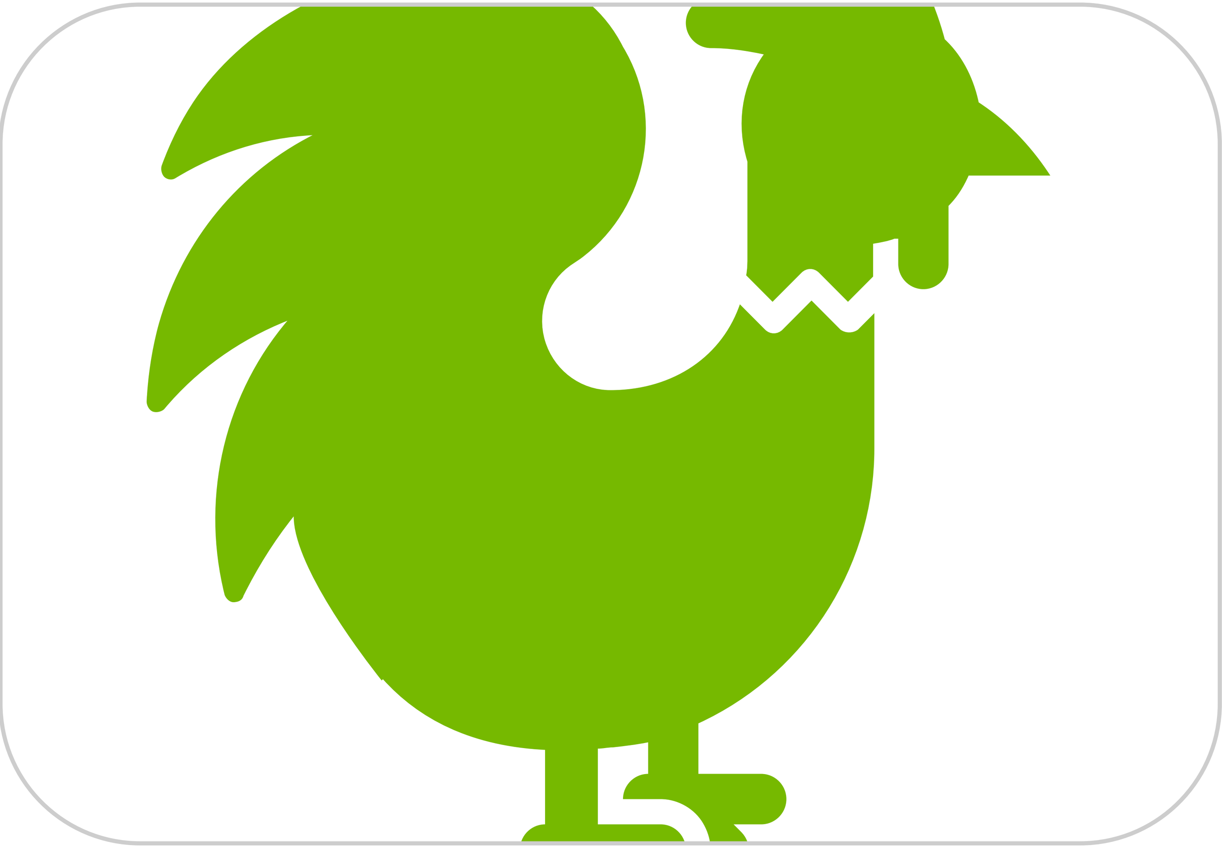
Intermediate Fusion



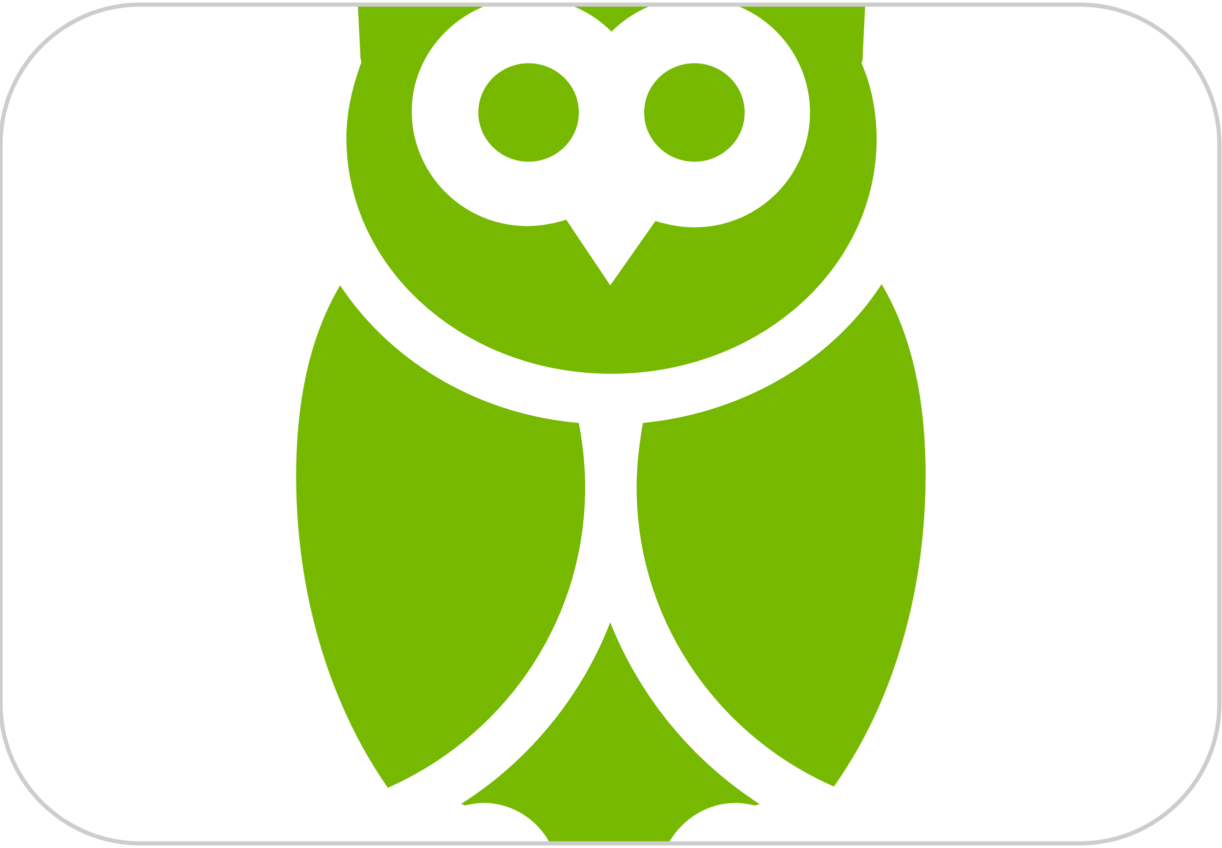
Multiplication Overfitting



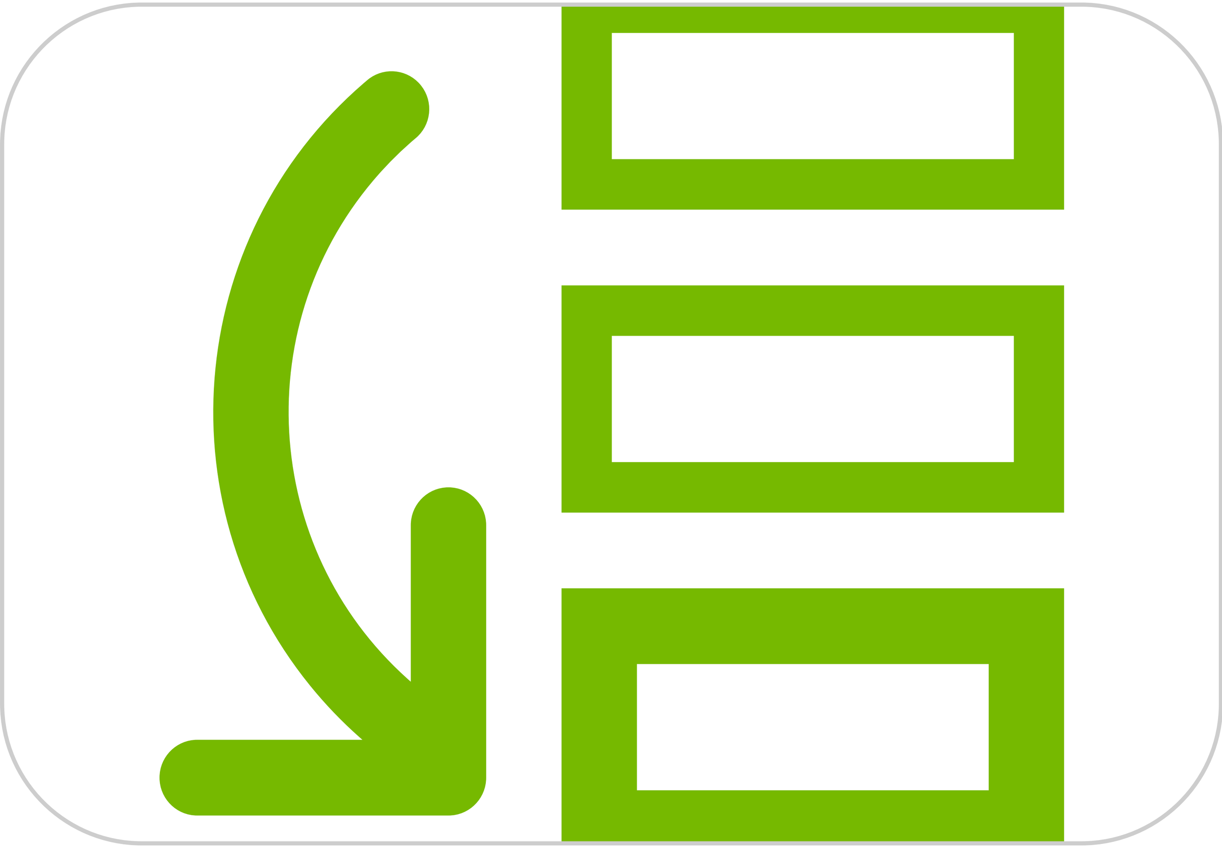
Lab Experiment



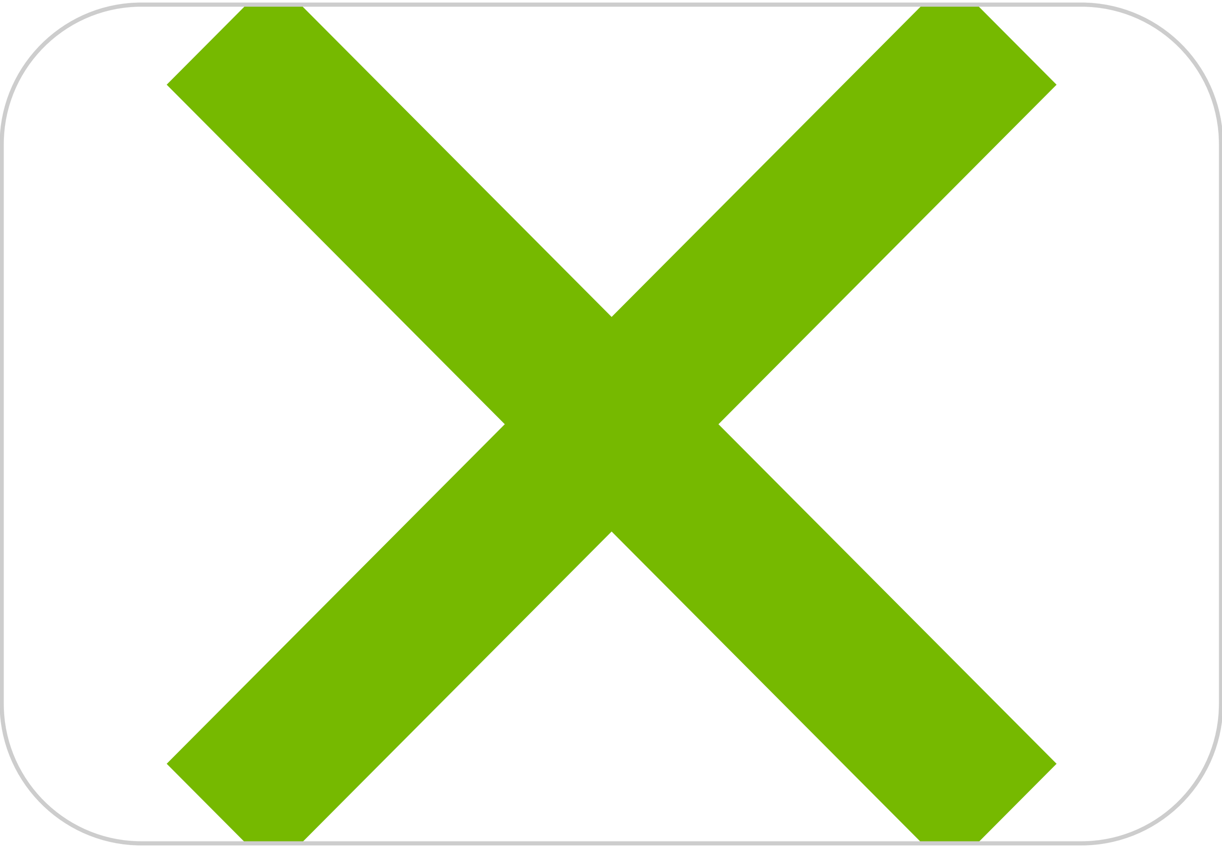
Early Fusion



Late Fusion



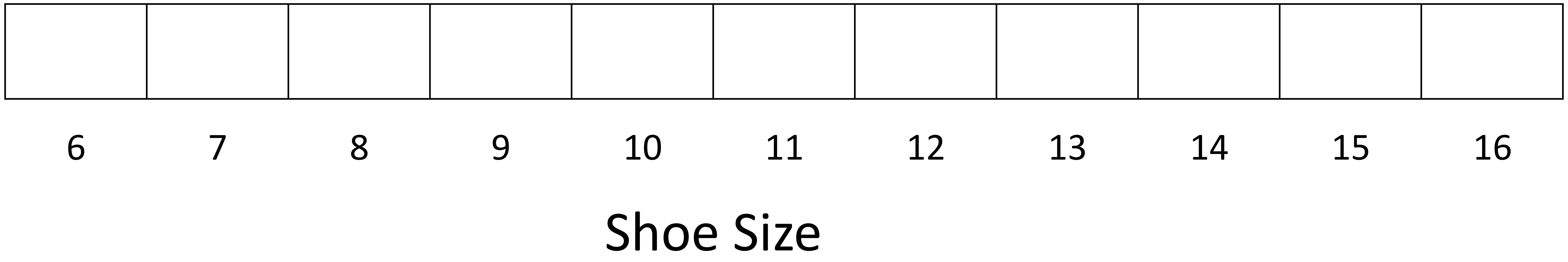
Intermediate Fusion
Concatenate



Intermediate Fusion
Multiplication

The Curse of Dimensionality

The Curse of Dimensionality



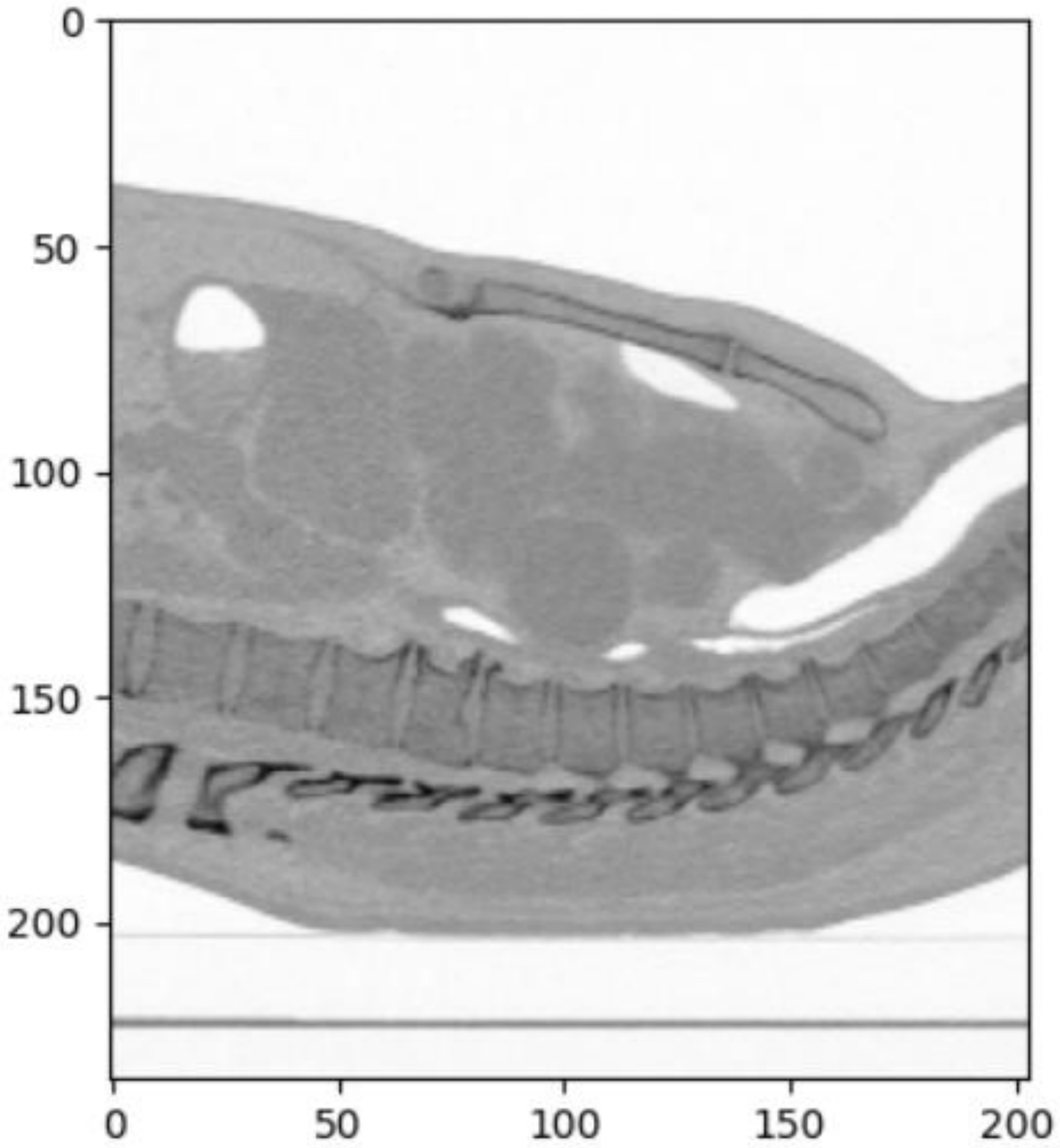
The Curse of Dimensionality

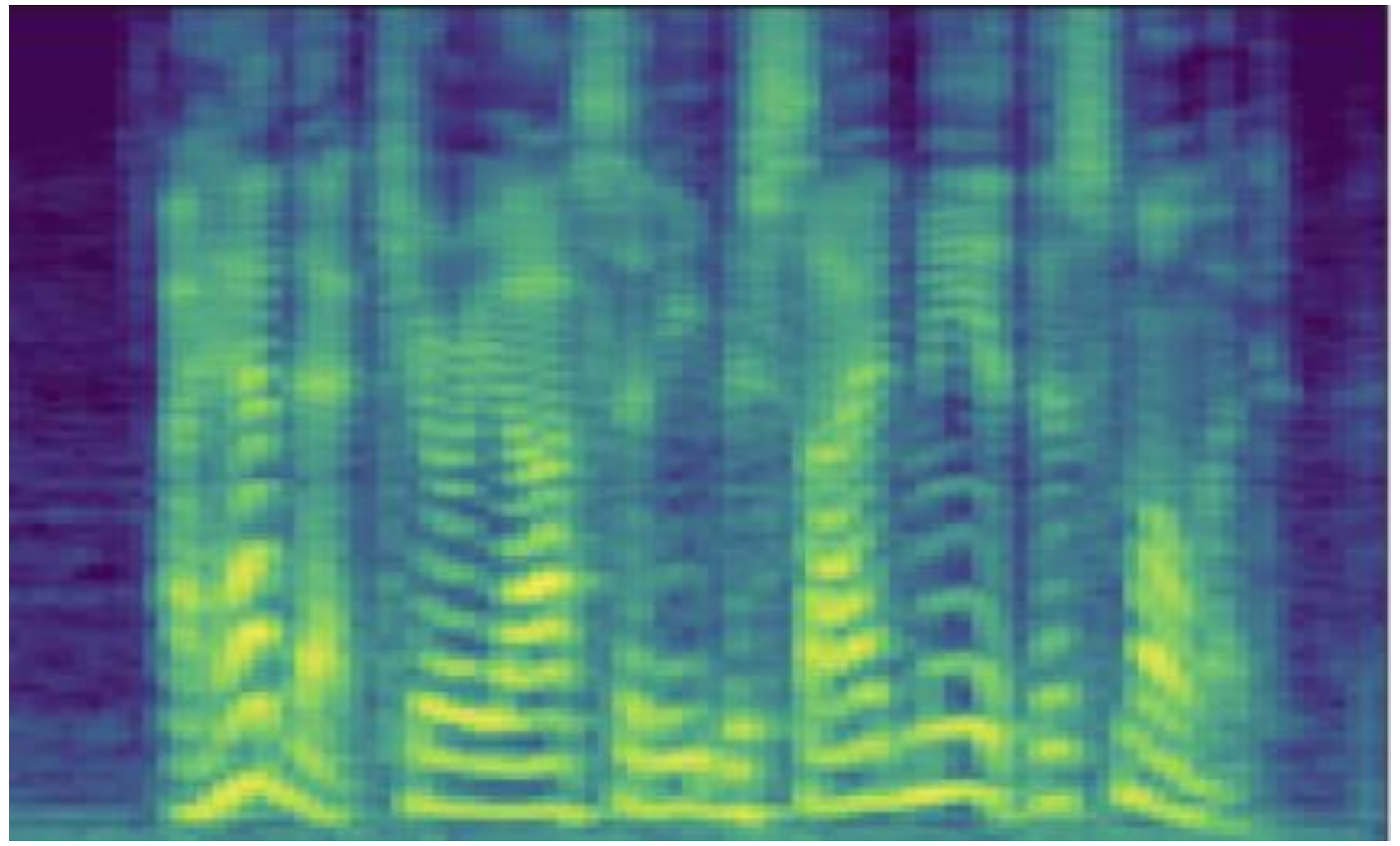
The Curse of Dimensionality

Choosing Data

Analysis Paralysis

CT Scan



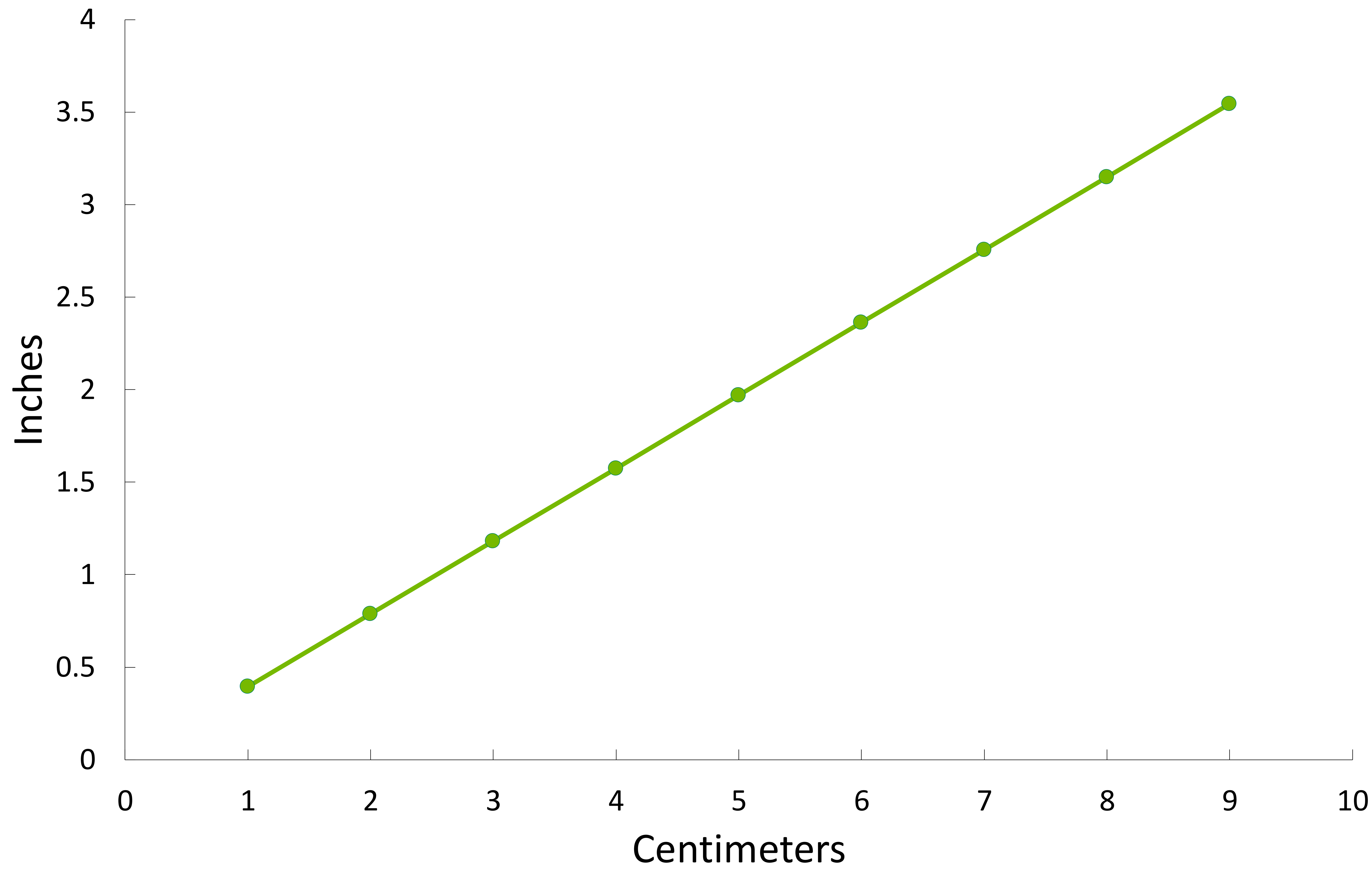


Spectrogram

Magnetic Data

Choosing Data

Highly Correlated Data



Lab 2b – Contrastive Pretraining

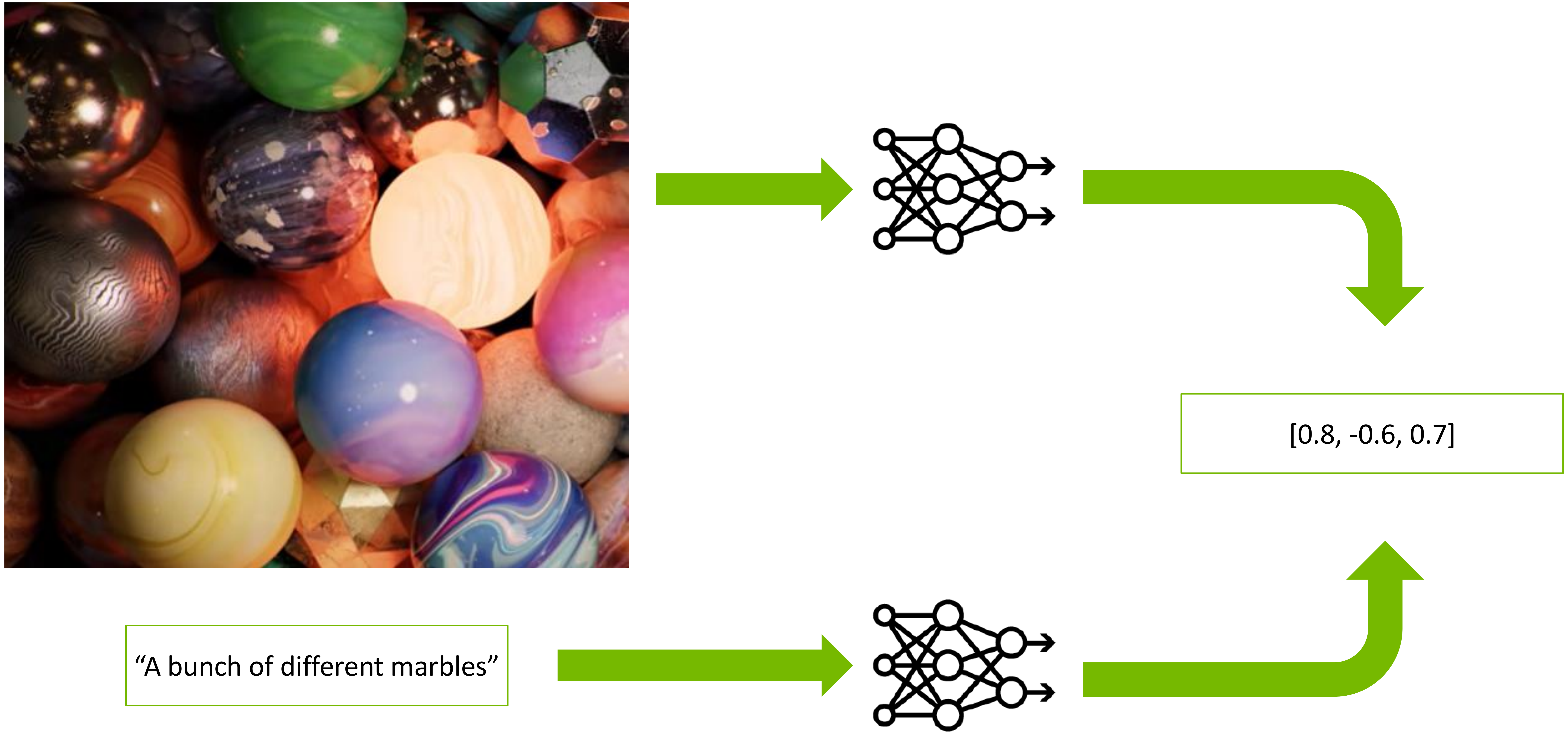
Matching Text to Image

Is it Possible?

"A bunch of different marbles"

Matching Text to Image

Is it Possible?



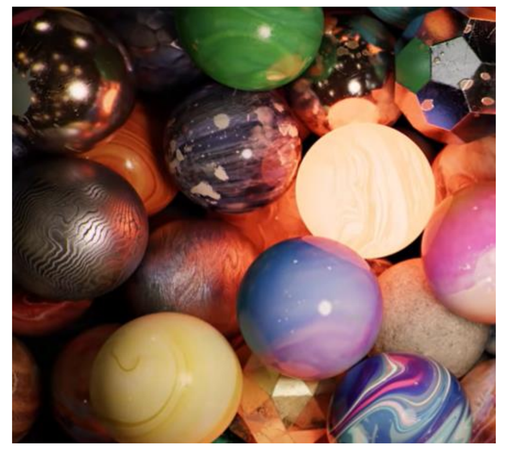
Cosine Similarity

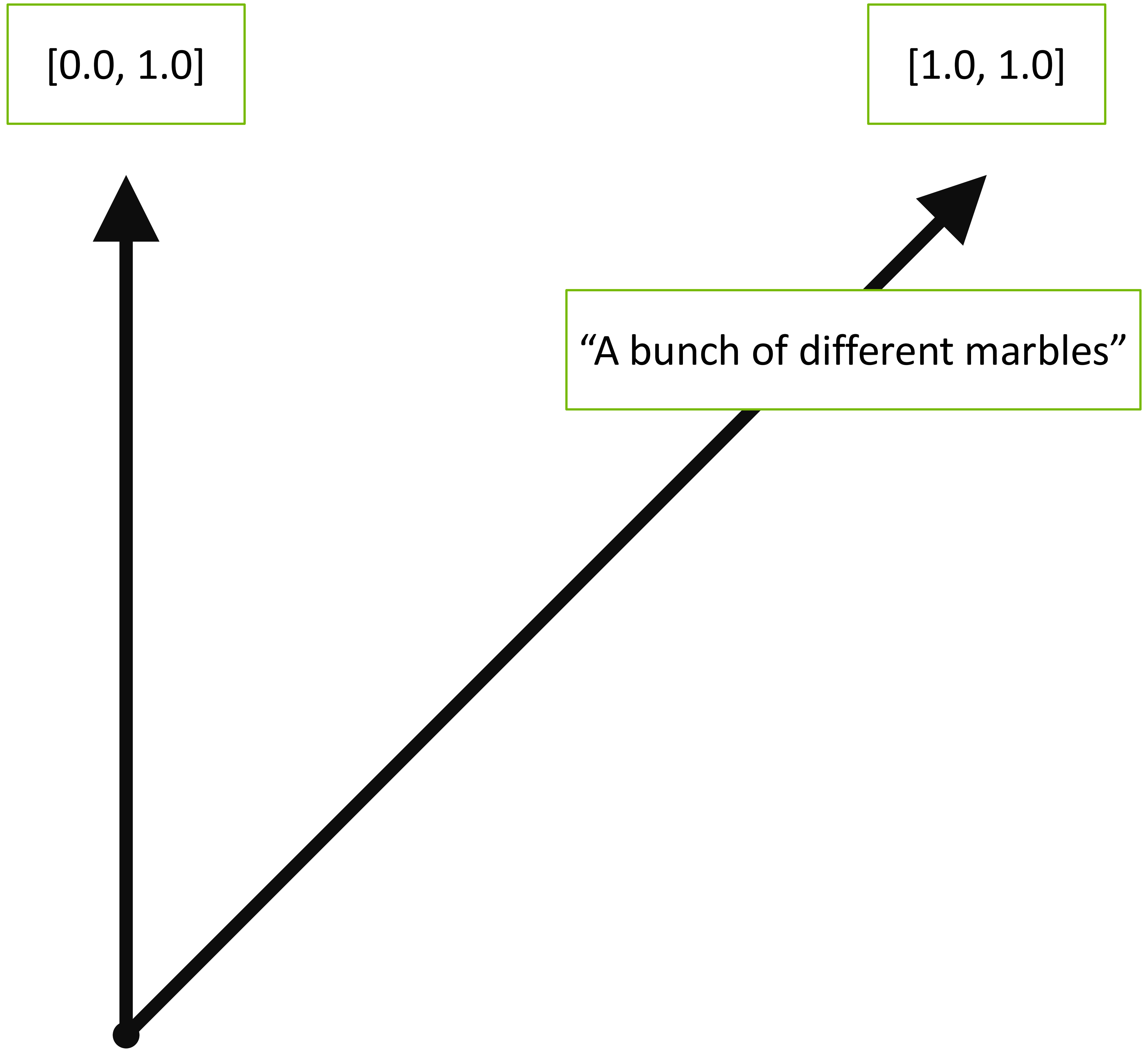
[0.0, 1.0]

“A bunch of different marbles”

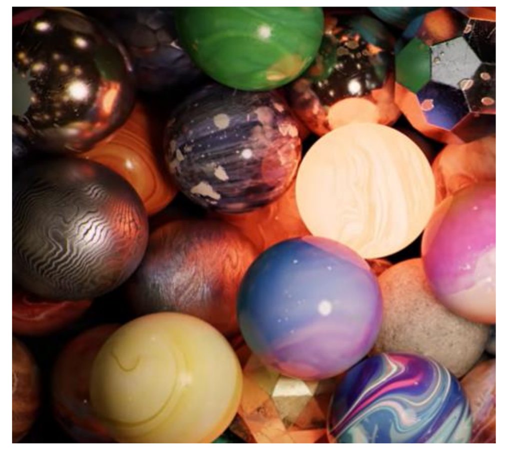
[1.0, 1.0]

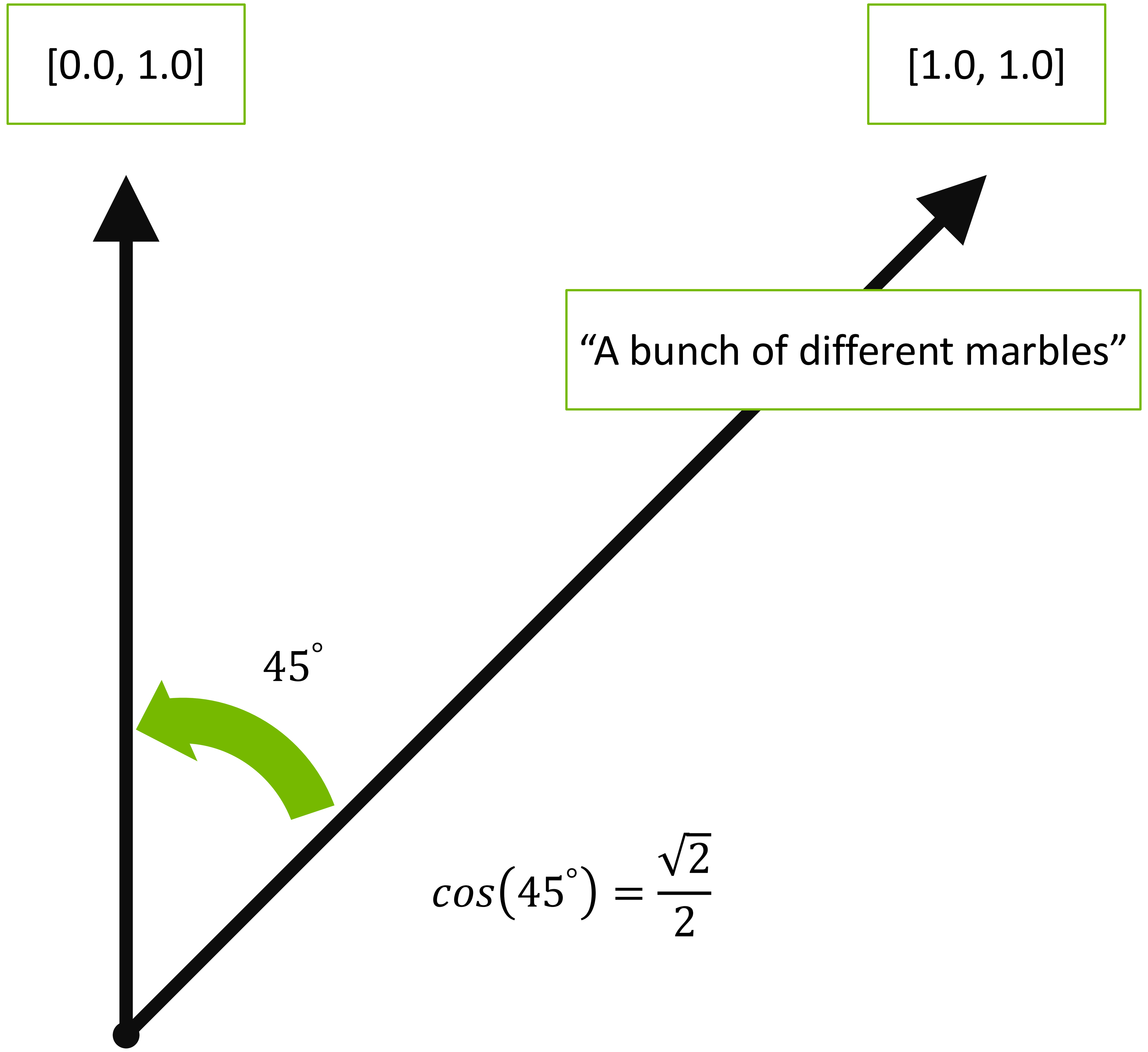
Cosine Similarity



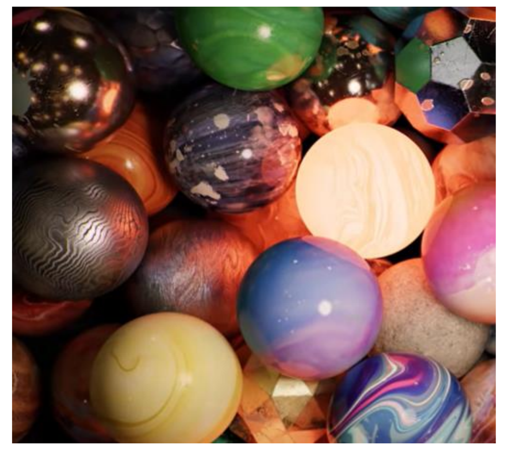


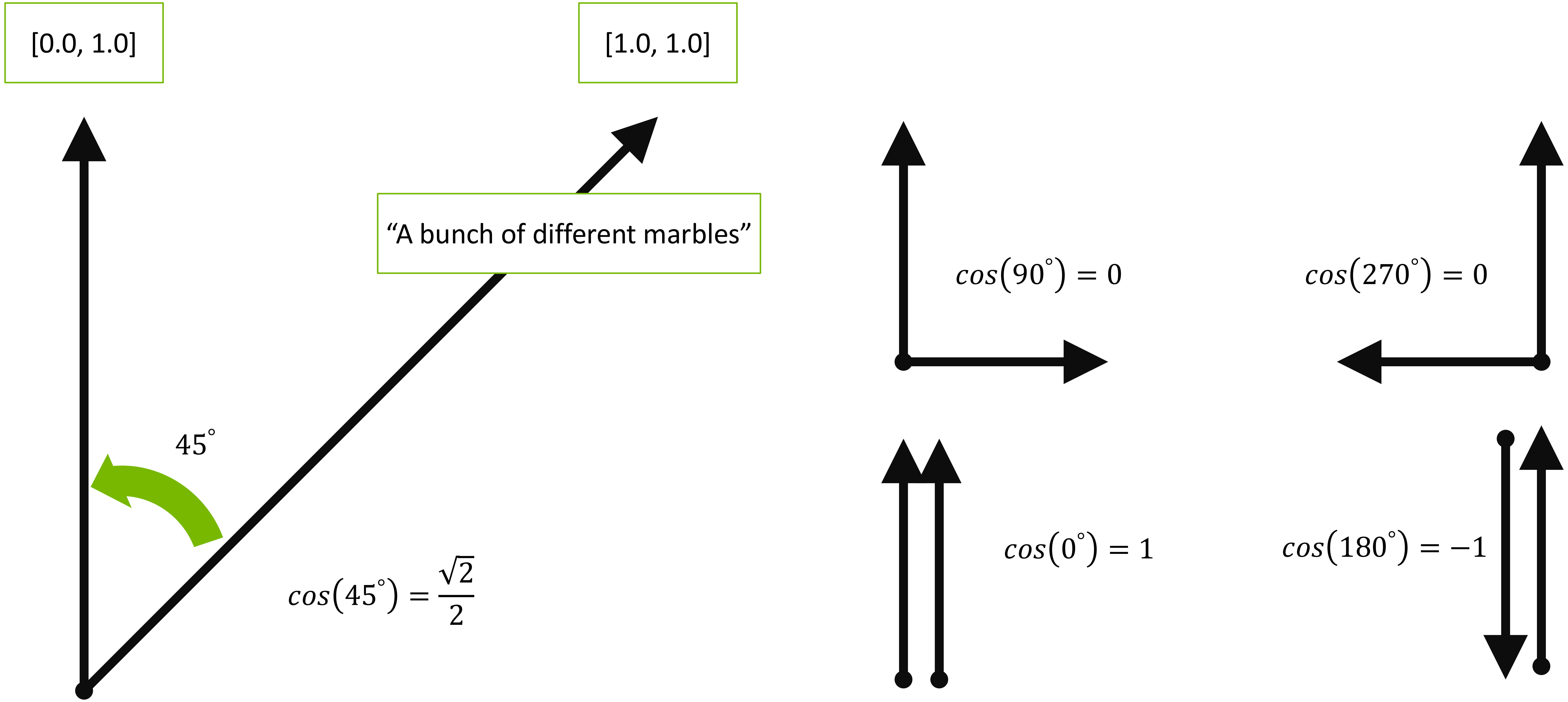
Cosine Similarity



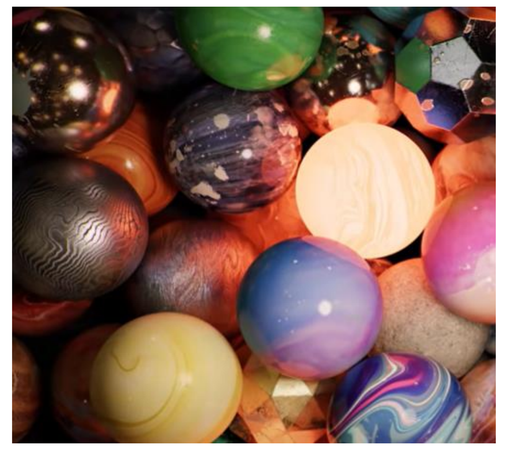


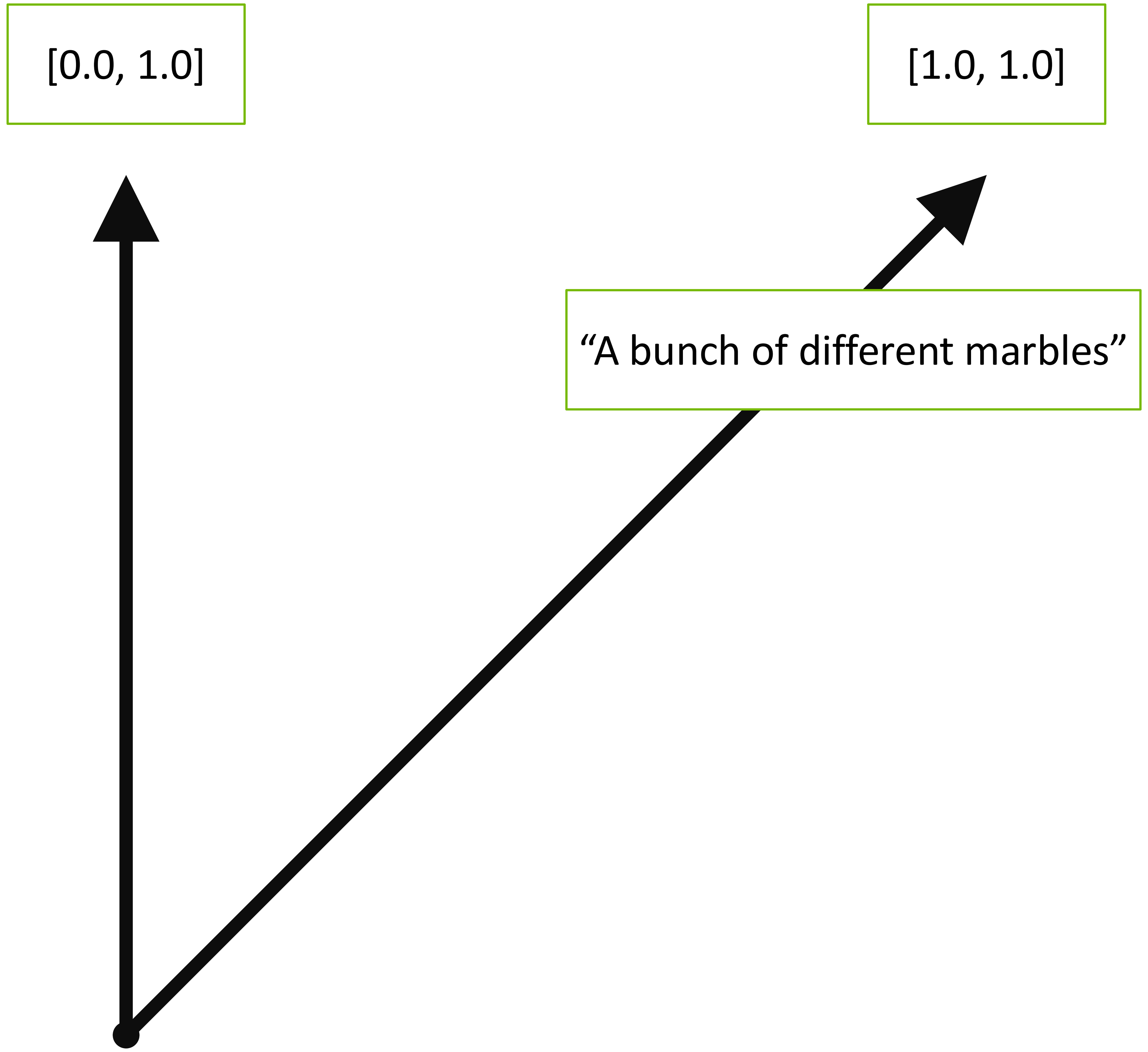
Cosine Similarity

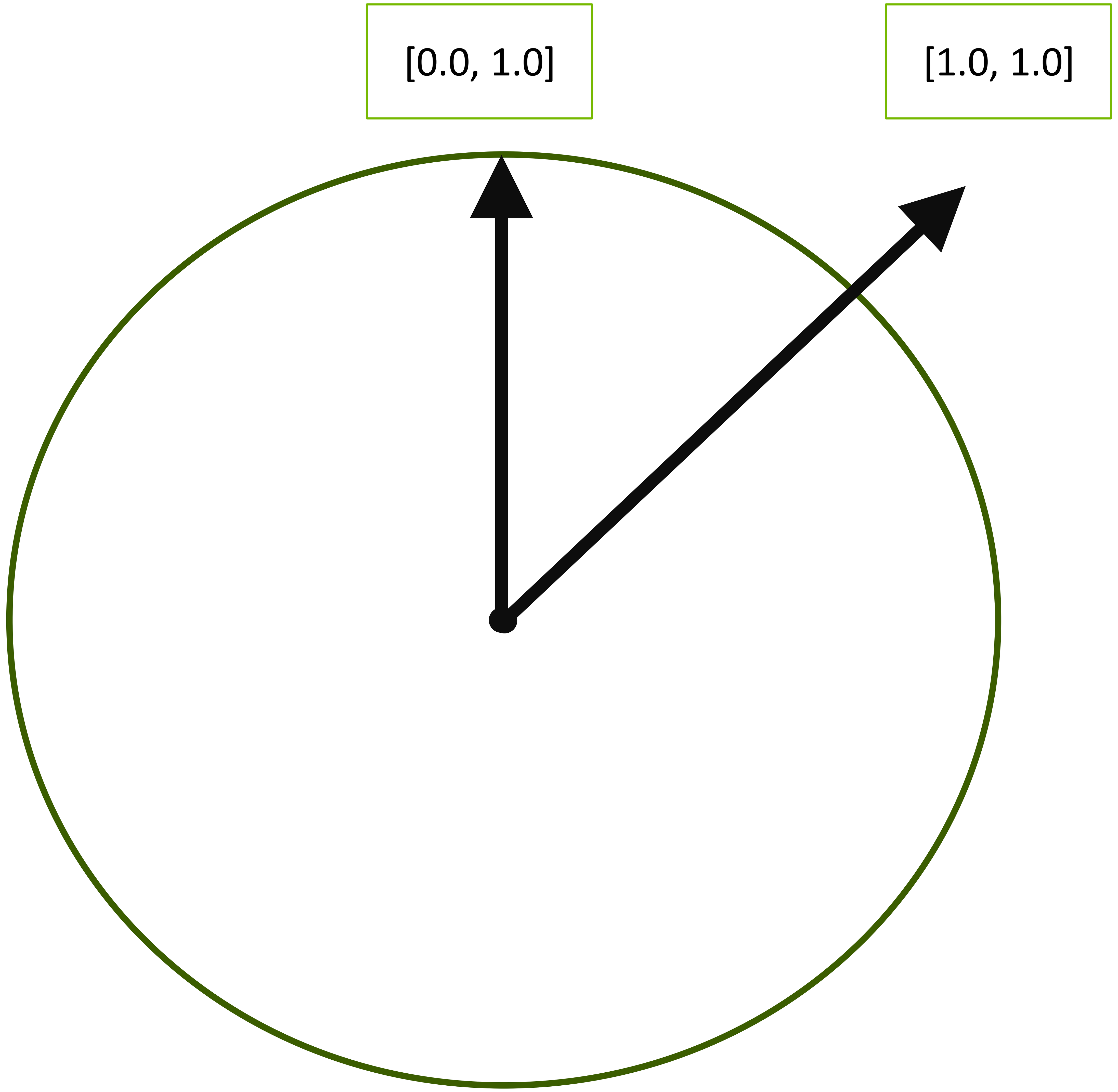




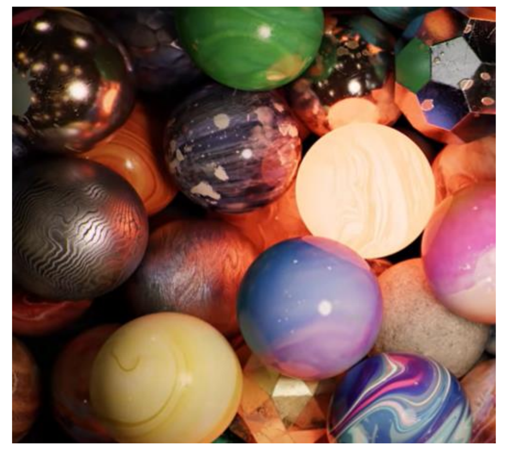
Dot Product

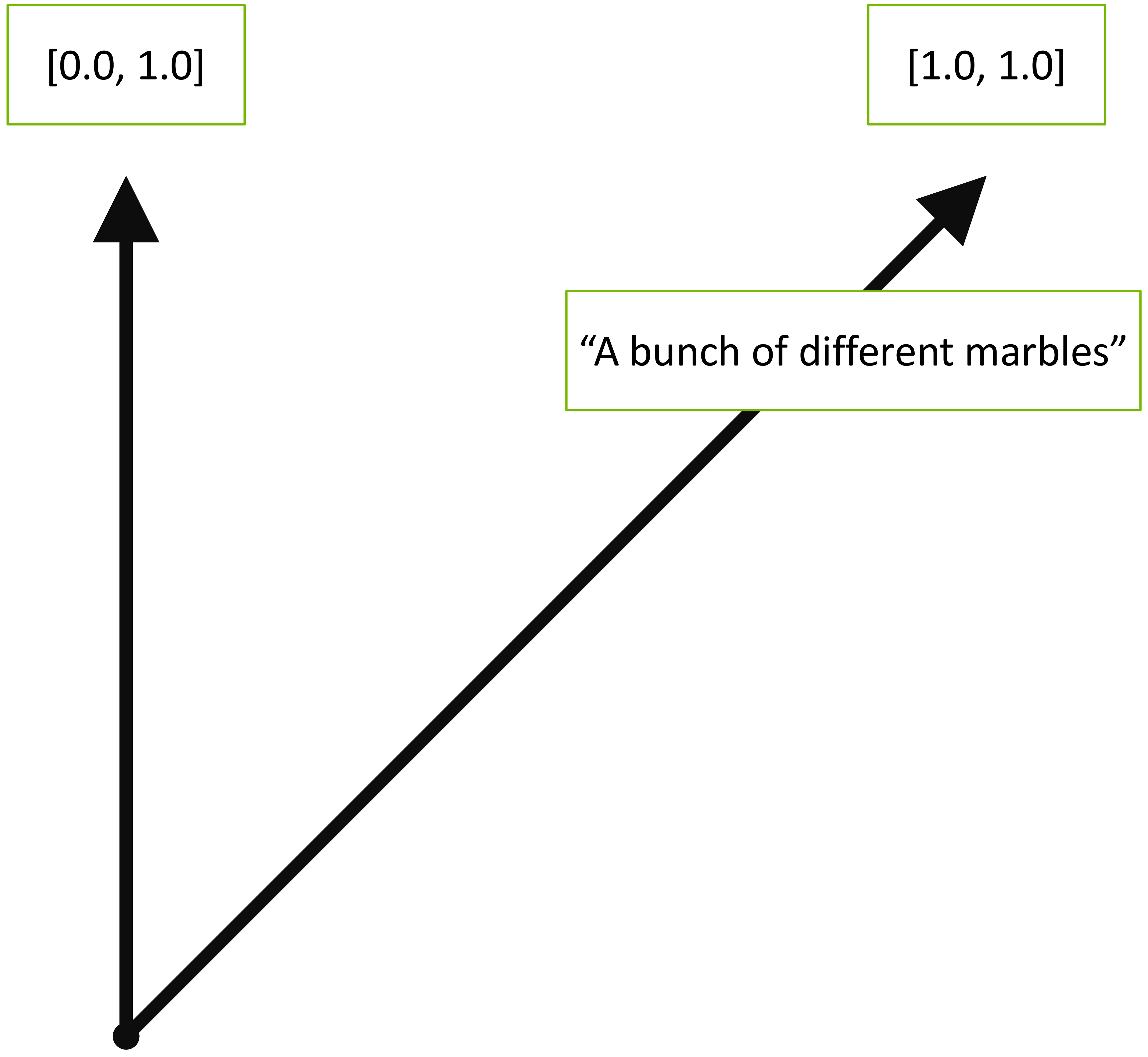


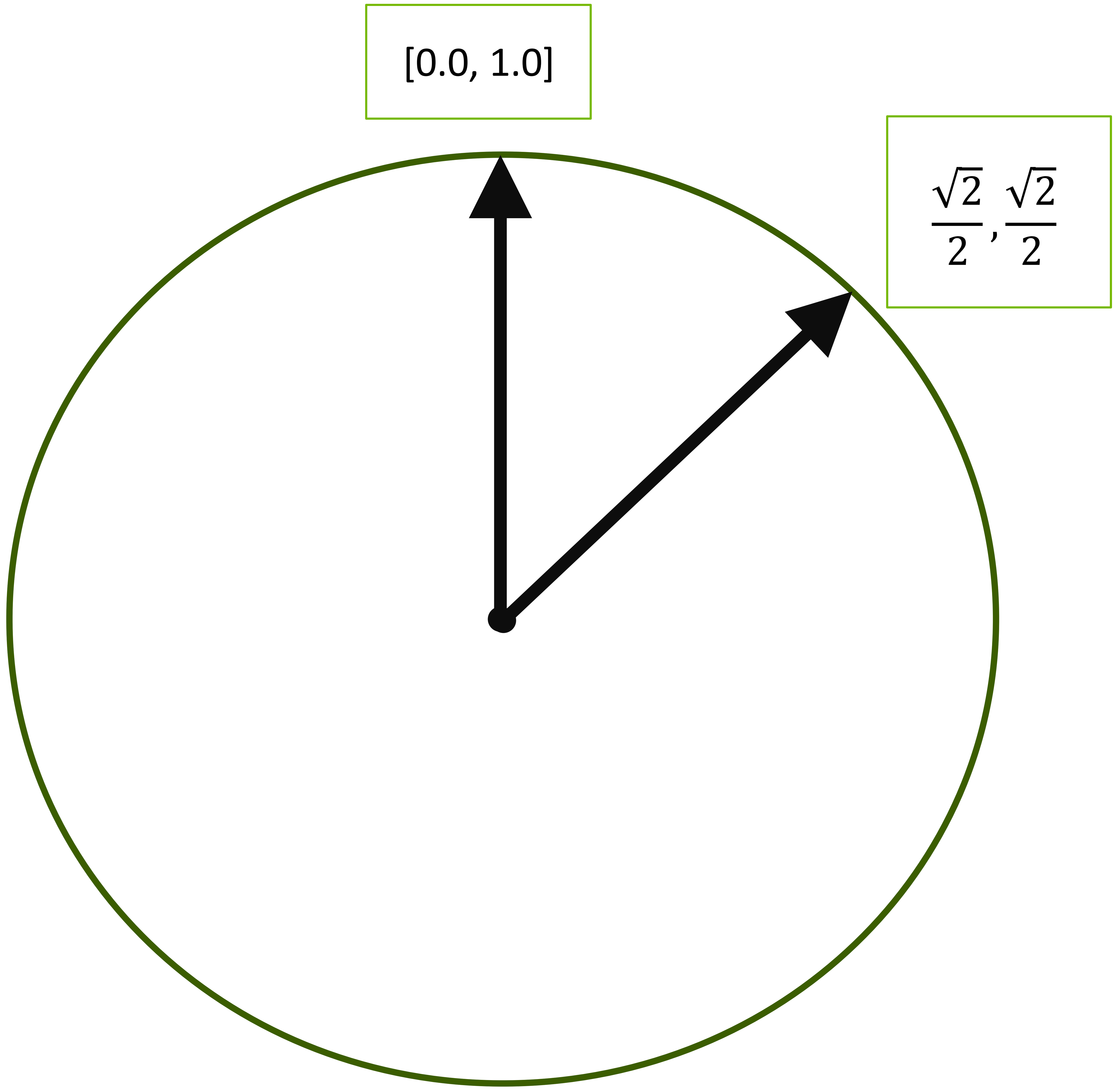




Dot Product



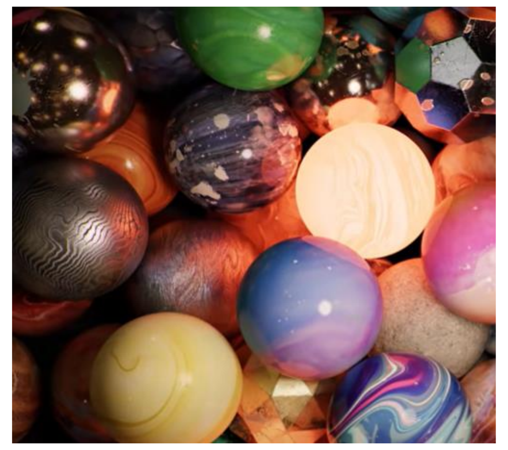

[0.0, 1.0]
[1.0, 1.0]
“A bunch of different marbles”

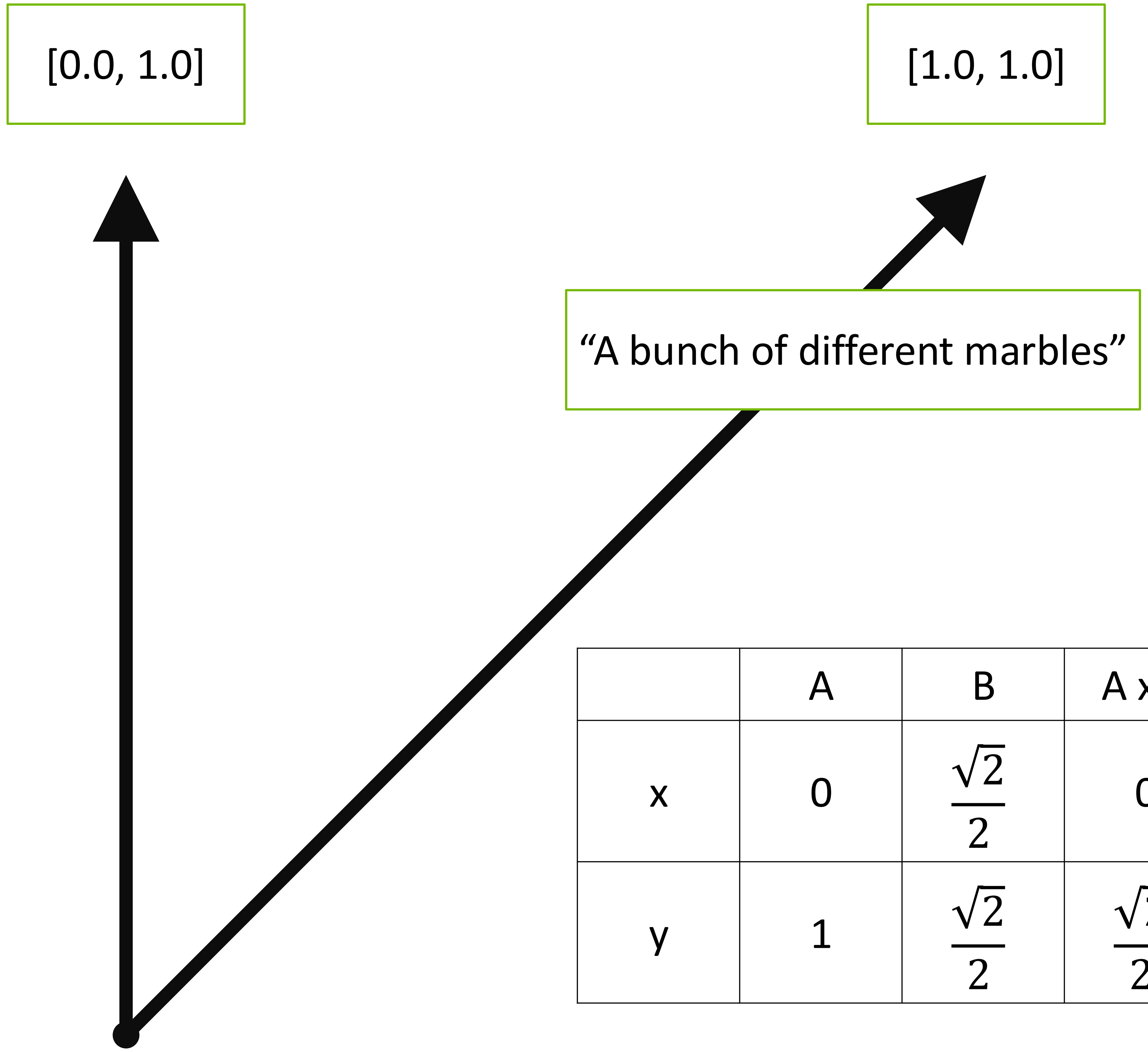

[0.0, 1.0]
$\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}$

Dot Product

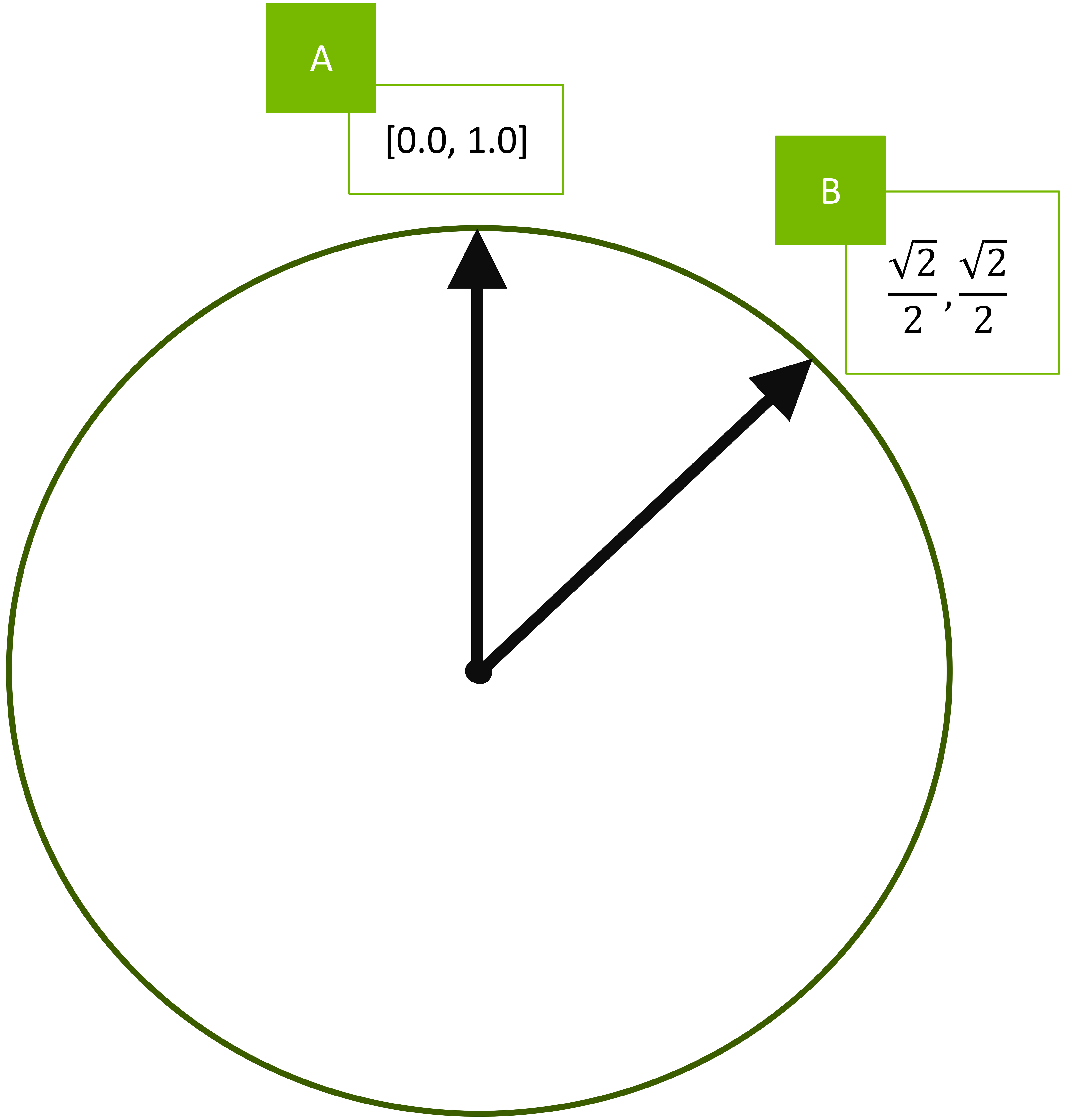

[0.0, 1.0]
[1.0, 1.0]
"A bunch of different marbles"
A
[0.0, 1.0]
B
$\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}$

Dot Product

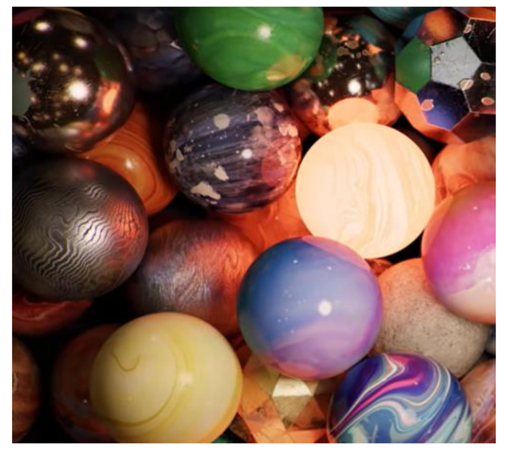


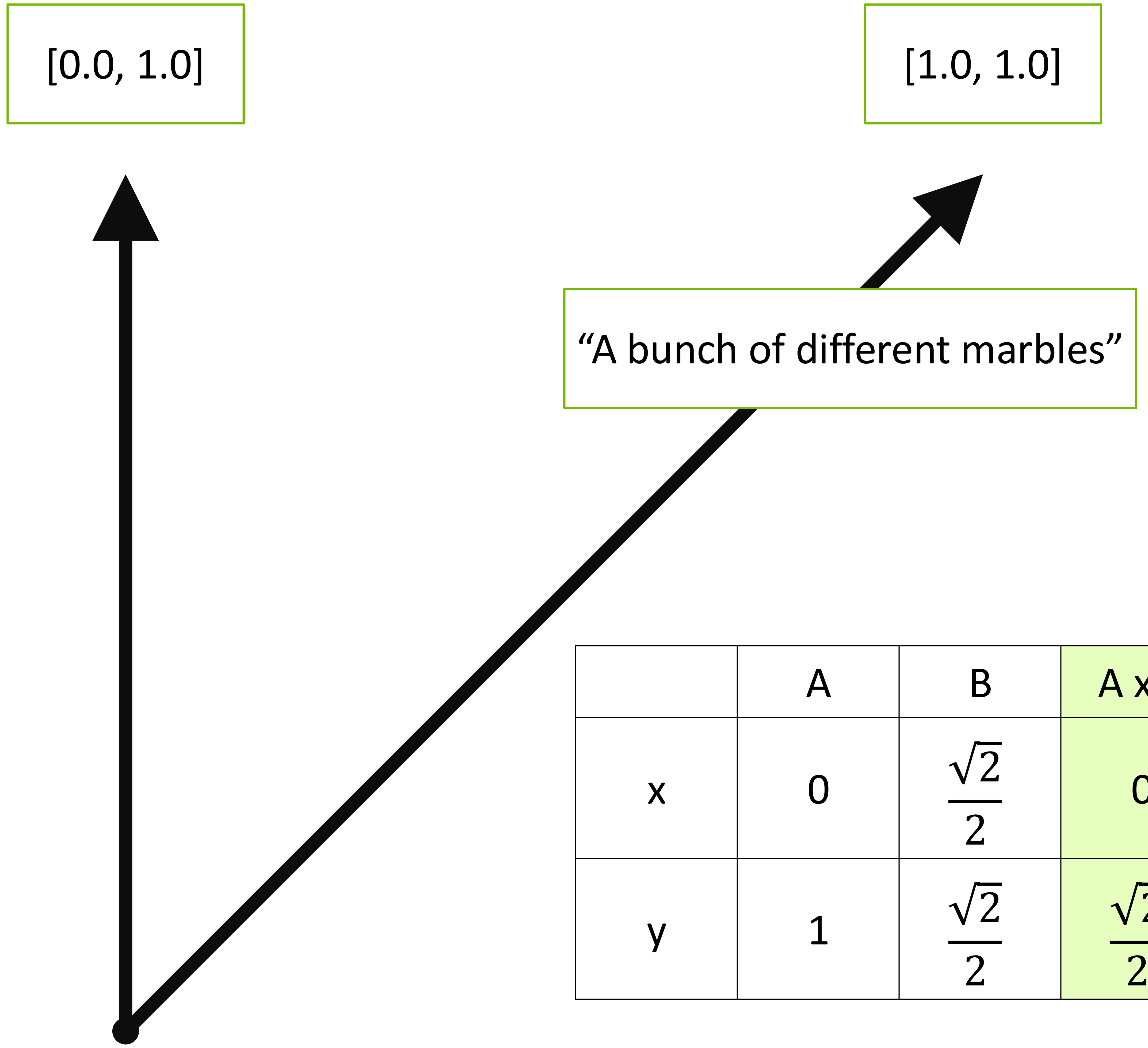


	A	B	A x B
x	0	$\frac{\sqrt{2}}{2}$	0
y	1	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$

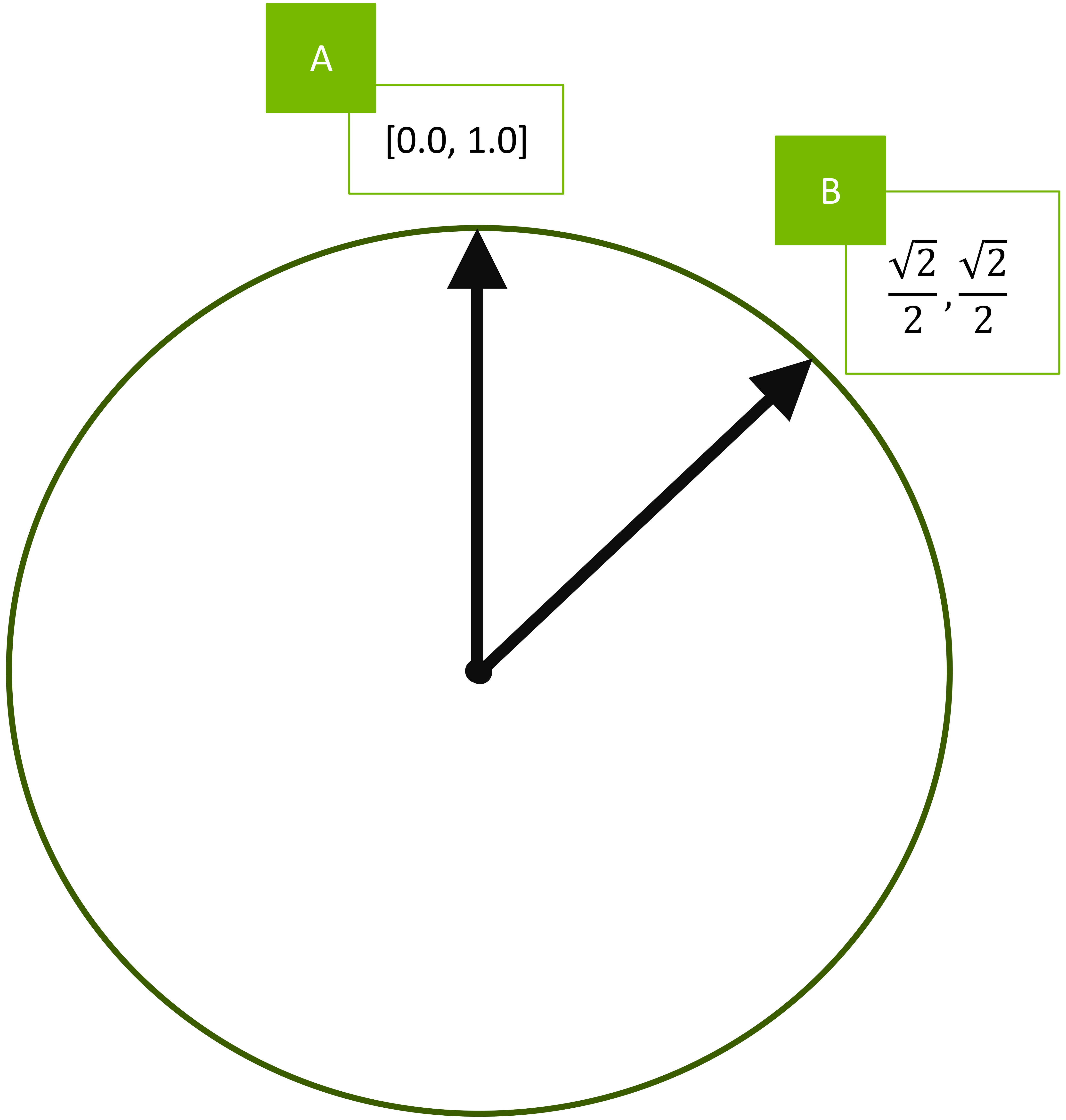


Dot Product

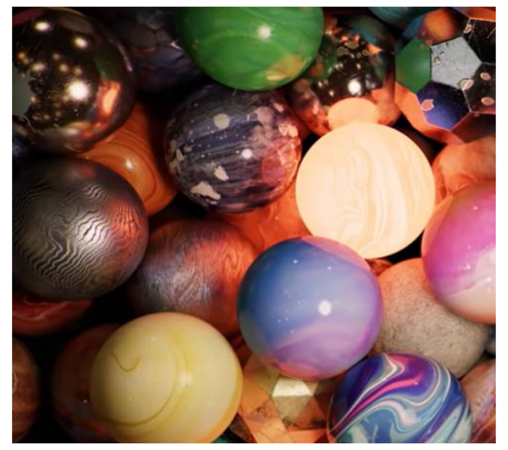




	A	B	A x B
x	0	$\frac{\sqrt{2}}{2}$	0
y	1	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$

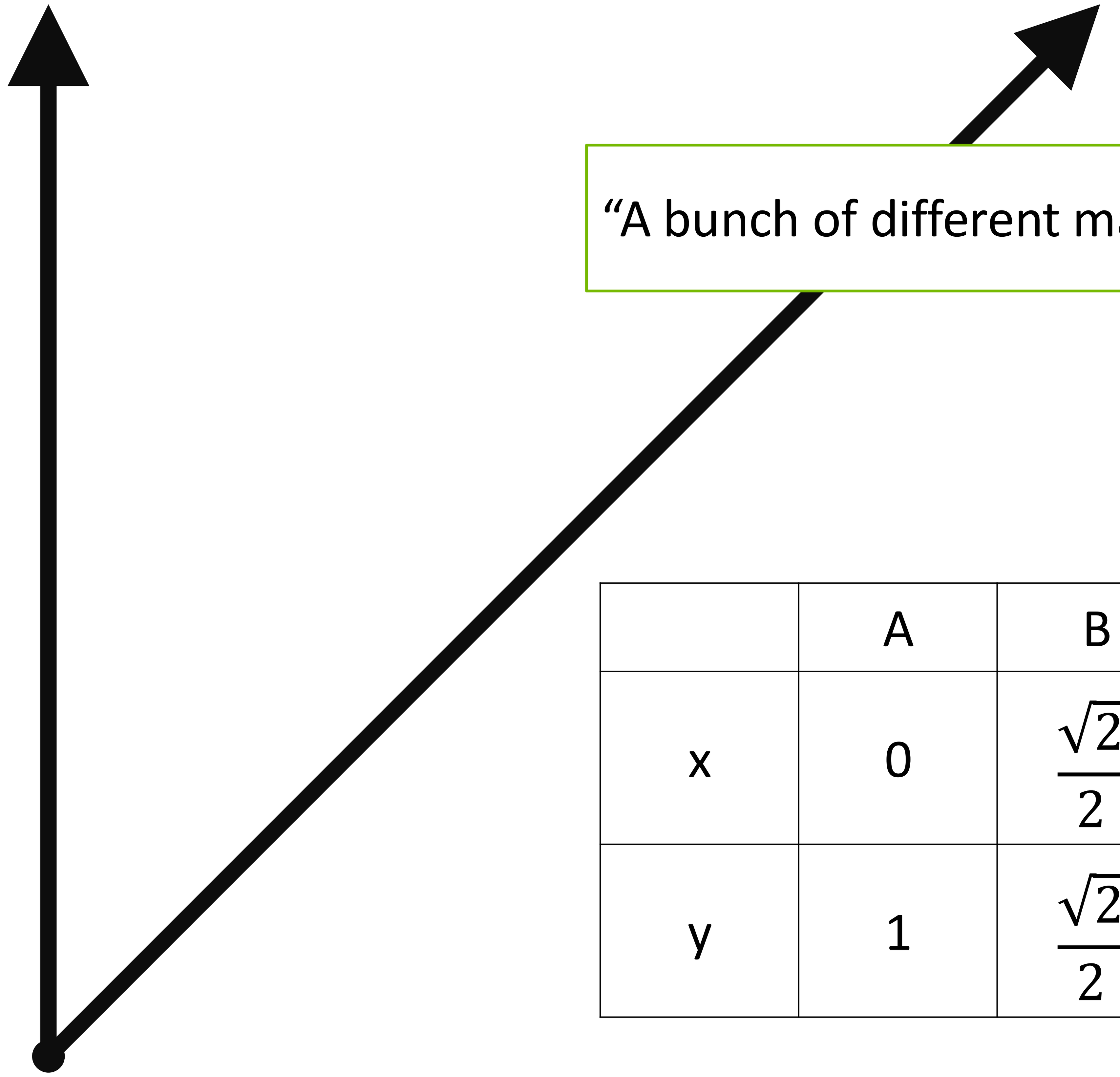


Dot Product



[0.0, 1.0]

[1.0, 1.0]



“A bunch of different marbles”

	A	B	A x B
x	0	$\frac{\sqrt{2}}{2}$	0
y	1	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$

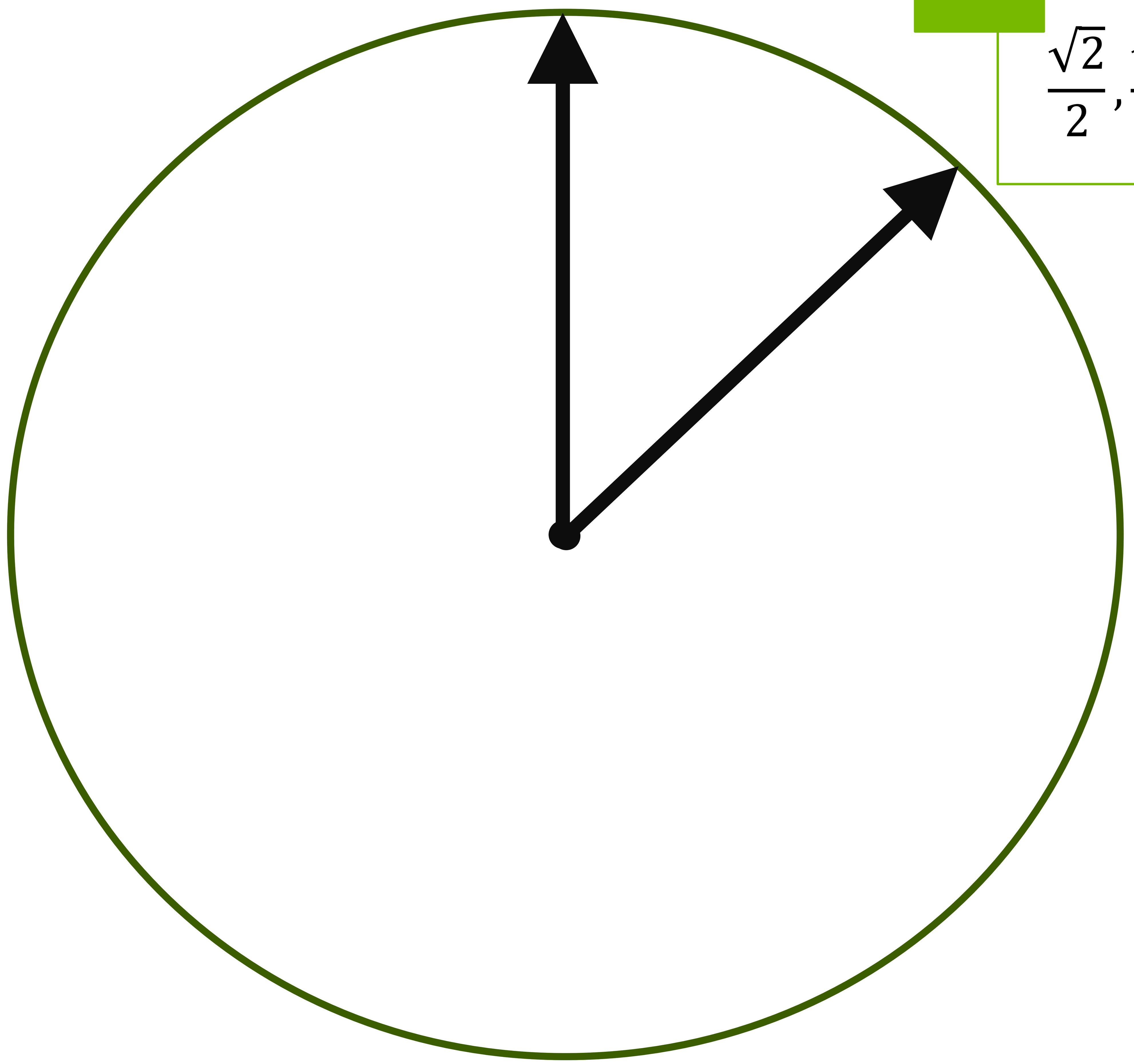
$$\frac{\sqrt{2}}{2}$$

A

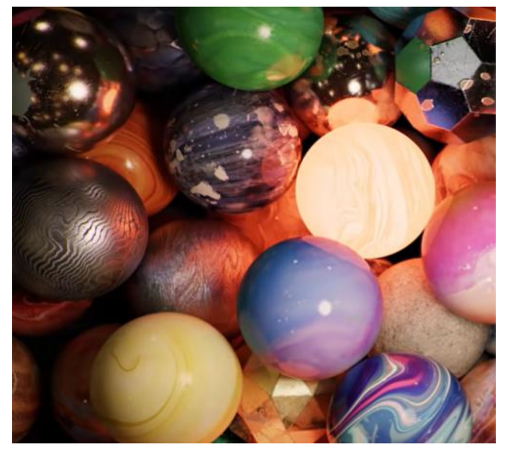
[0.0, 1.0]

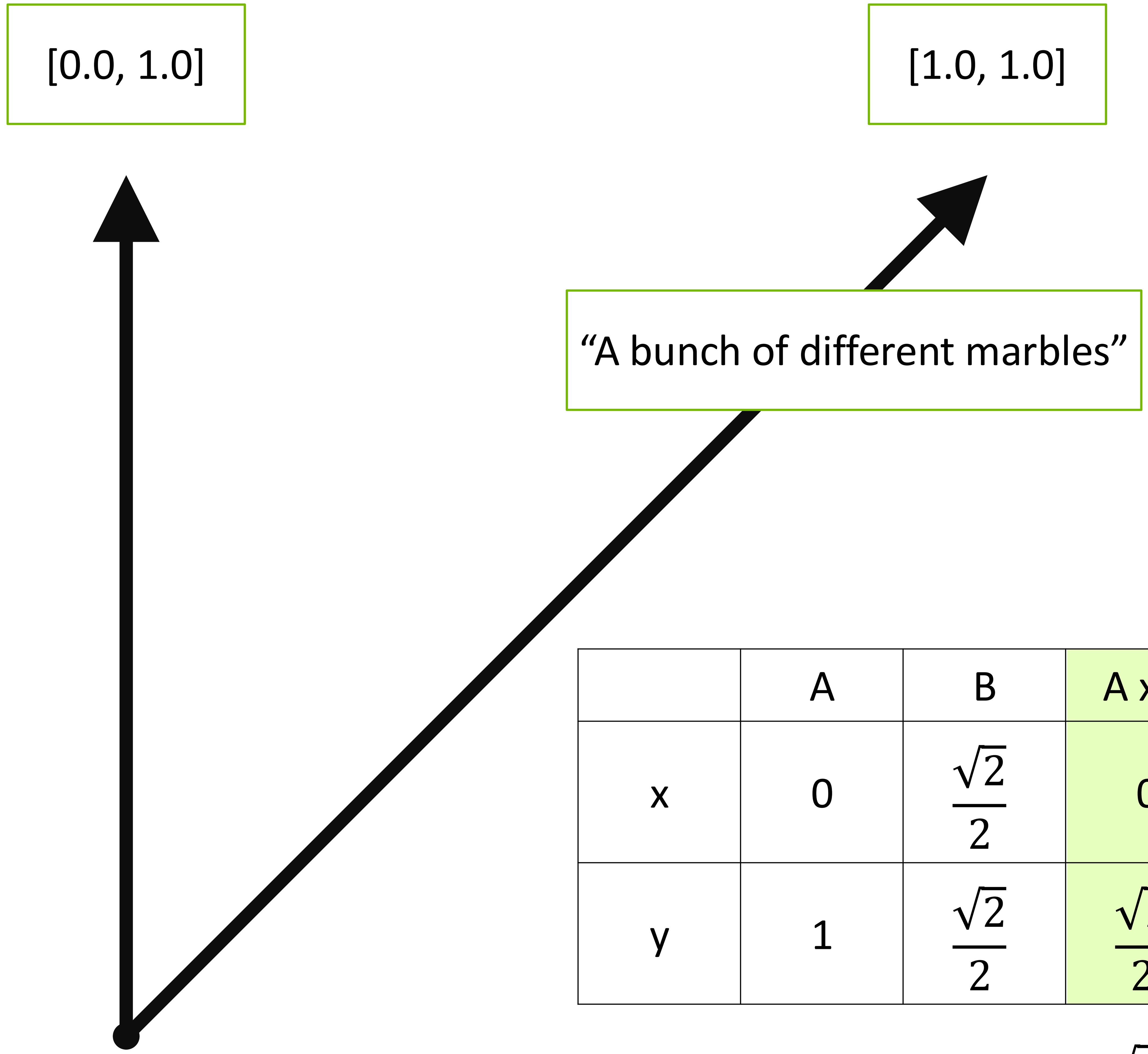
B

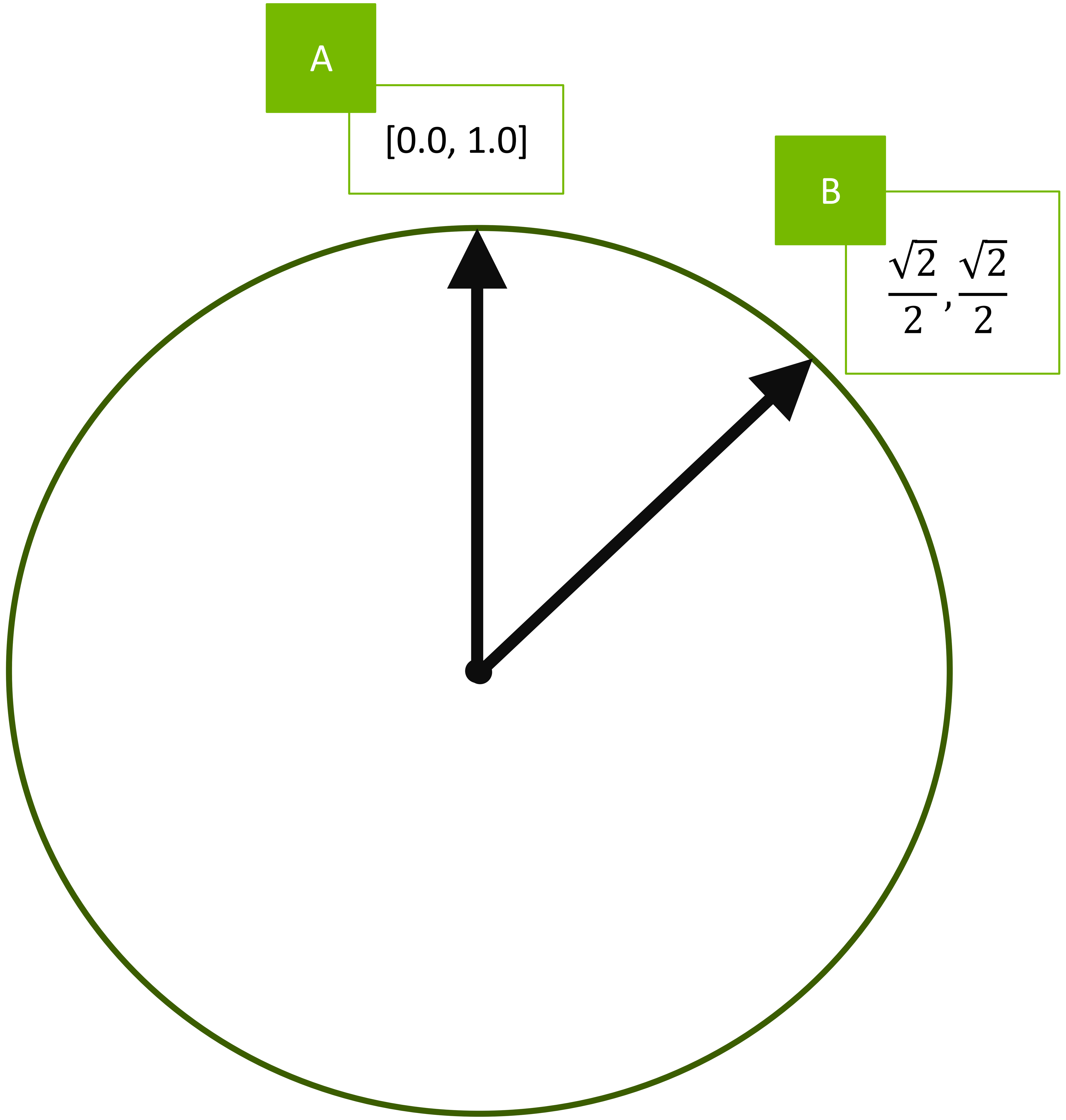
$\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}$



Dot Product







	A	B	A x B
x	0	$\frac{\sqrt{2}}{2}$	0
y	1	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$

$$cos(45^{\circ}) = \frac{\sqrt{2}}{2}$$

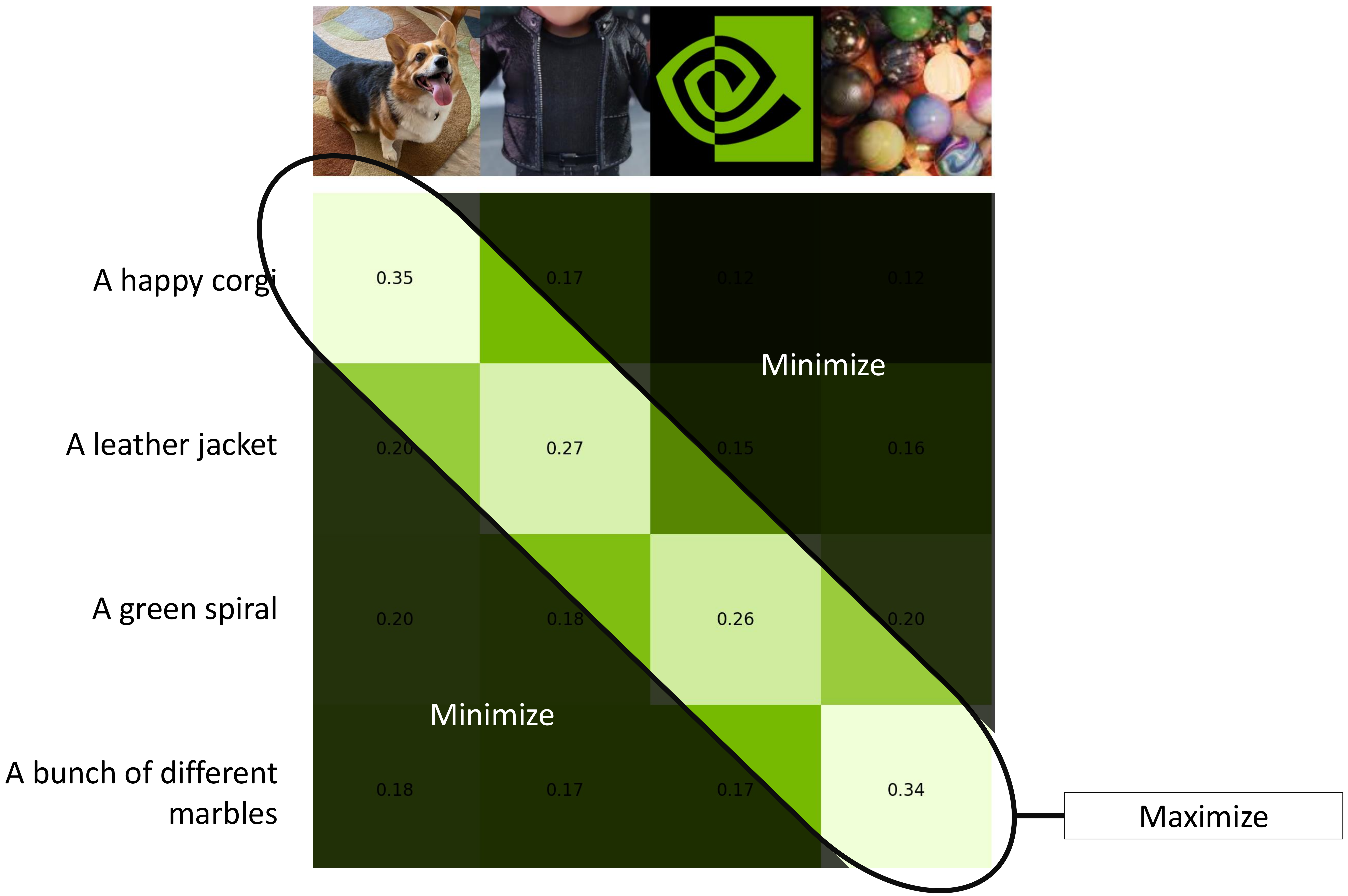
CLIP Training

A happy corgi	0.35	0.17	0.12	0.12
A leather jacket	0.20	0.27	0.15	0.16
A green spiral	0.20	0.18	0.26	0.20
A bunch of different marbles	0.18	0.17	0.17	0.34

Cosine similarity between encoding for "A happy corgi" and encoding for each image

Cosine similarity between encoding for the NVIDIA logo and encoding for each text

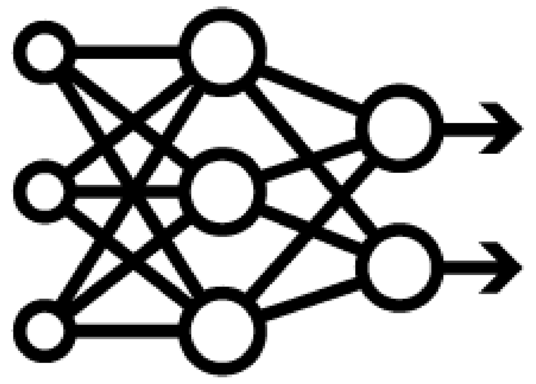
CLIP Training



RAG and Vector Databases

Vector Databases

CLIP



[0.7, 0.2, -0.1]

[-0.1, 0.1, -0.8]

[-0.2, -0.1, 0.4]

[0.8, -0.6, 0.7]

Retrieval Augmented Generation (RAG)

Retrieval Augmented Generation (RAG)

a cute corgi

[0.6, 0.1, -0.1]

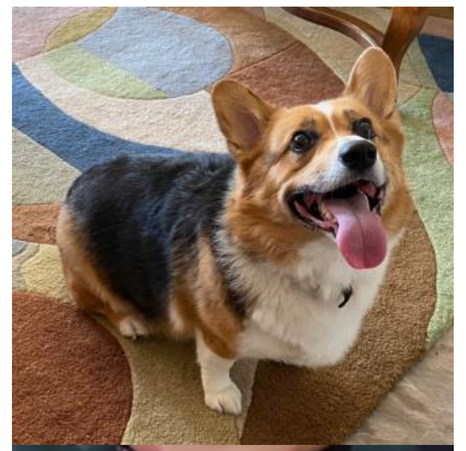
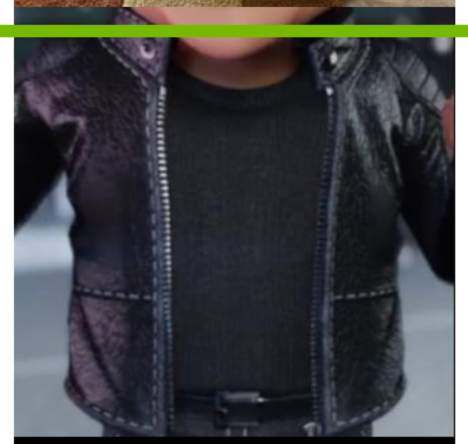
[0.7, 0.2, -0.1]	0.9933
[-0.1, 0.1, -0.8]	0.0599
[-0.2, -0.1, 0.4]	-0.6017
[0.8, -0.6, 0.7]	0.4651

Cosine Similarity

Retrieval Augmented Generation (RAG)

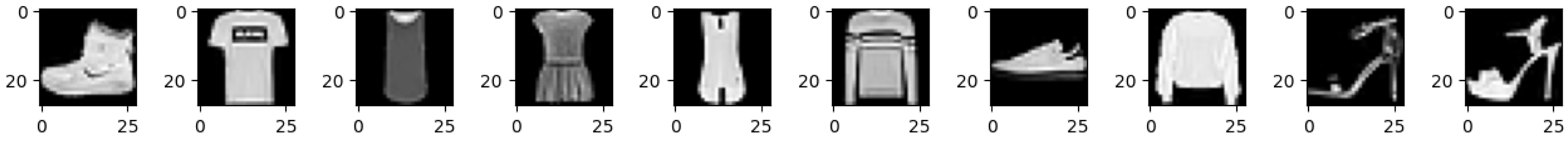
a cute corgi

[0.6, 0.1, -0.1]

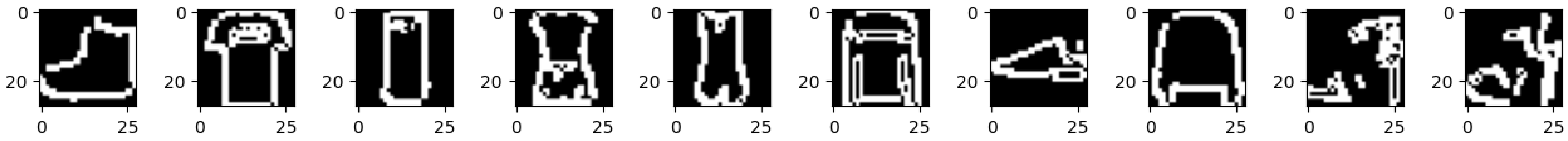
	[0.7, 0.2, -0.1]	**0.9933**	Best Match
	[-0.1, 0.1, -0.8]	0.0599	
	[-0.2, -0.1, 0.4]	-0.6017	
	[0.8, -0.6, 0.7]	0.4651	

Contrastive Outline-Image Pretraining

Image



Outline



Let’s Get Started!

NVIDIA